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Noon et al.

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(54) **ONE-SHOT CHALLENGE TO SEARCH AND ACCESS UNREDACTED VAULTED ELECTRONIC COMMUNICATIONS**

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Primary Examiner — Trang T Doan

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(57) **ABSTRACT**

Related U.S. Application Data

(60) Provisional application No. 63/303,369, filed on Jan. 26, 2022.

A tool is disclosed for searching information redacted from electronic communications. The tool receives, based on input by a user into an interface of an electronic communication repository, a request to search through vaulted information, the vaulted information redacted from the electronic communication repository. The tool transmits an identity verification challenge to a device of the user and determines whether the identity verification challenge is successful. Responsive to determining that the identity verification challenge is successful, the tool searches the vaulted information for one or more files comprising a symbol input by the user, and generates, for display within the interface, identifiers corresponding to each of the one or more files.

(51) **Int. Cl.**
G06F 21/62 (2013.01)

G06F 21/60 (2013.01)

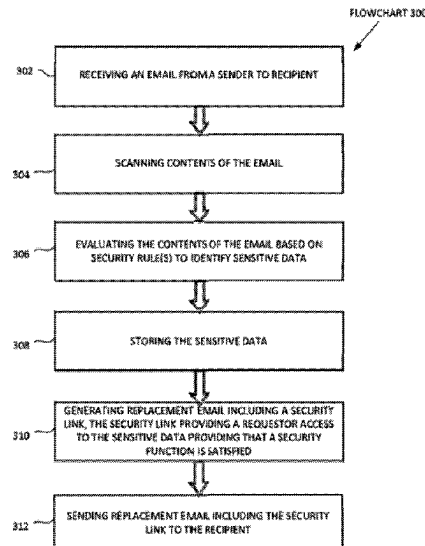
(52) **U.S. Cl.**
CPC **G06F 21/6254** (2013.01); **G06F 21/604** (2013.01); **G06F 2221/2103** (2013.01); **G06F 2221/2113** (2013.01); **G06F 2221/2141** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

20 Claims, 21 Drawing Sheets



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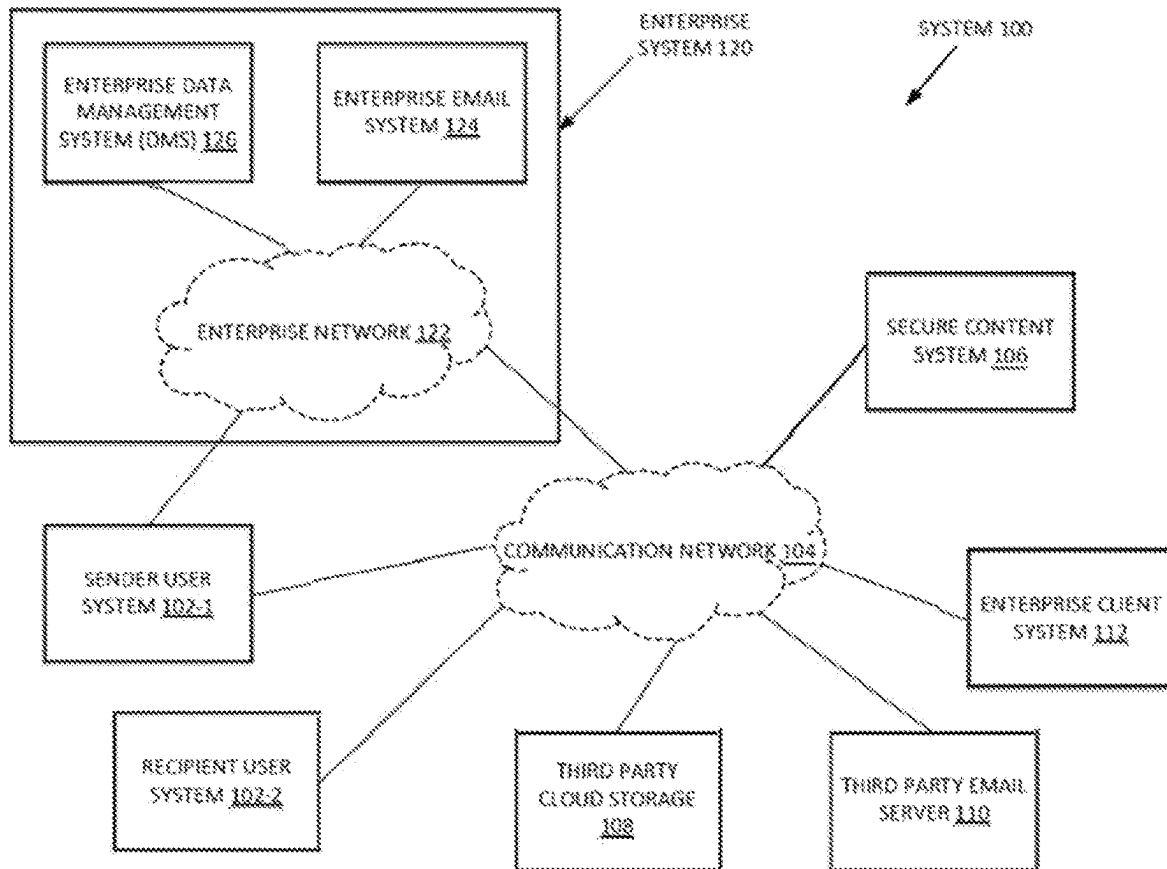
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**FIG. 1**

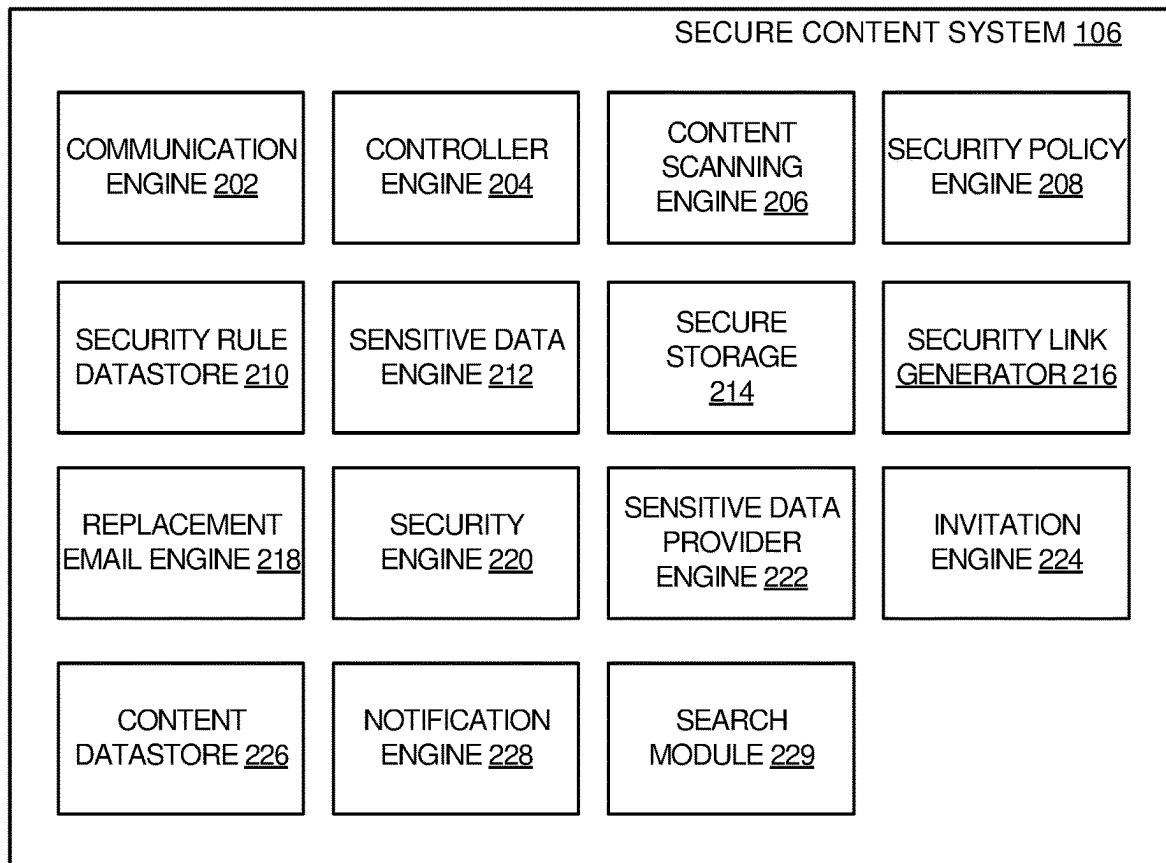


FIG. 2

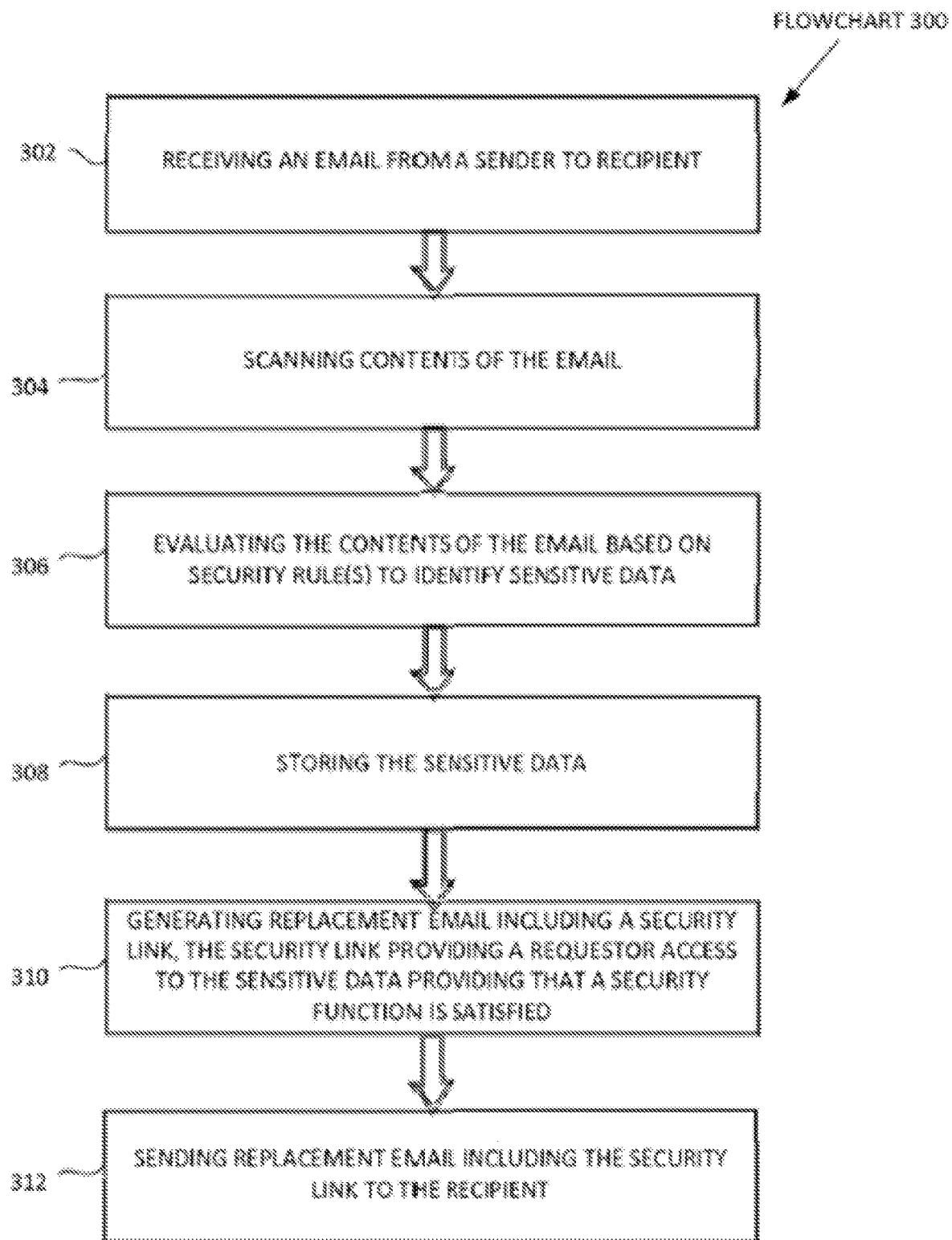


FIG. 3

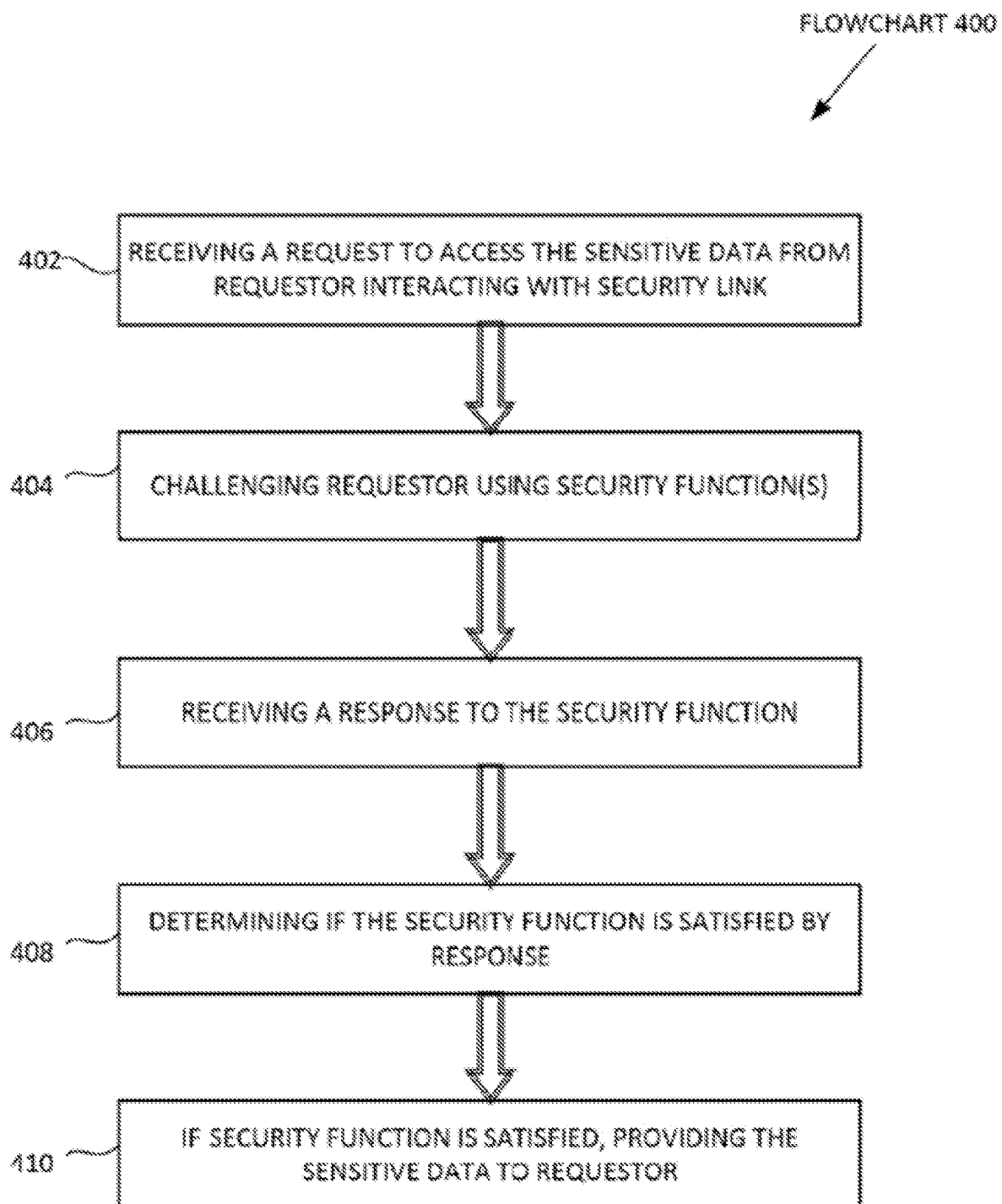


FIG. 4

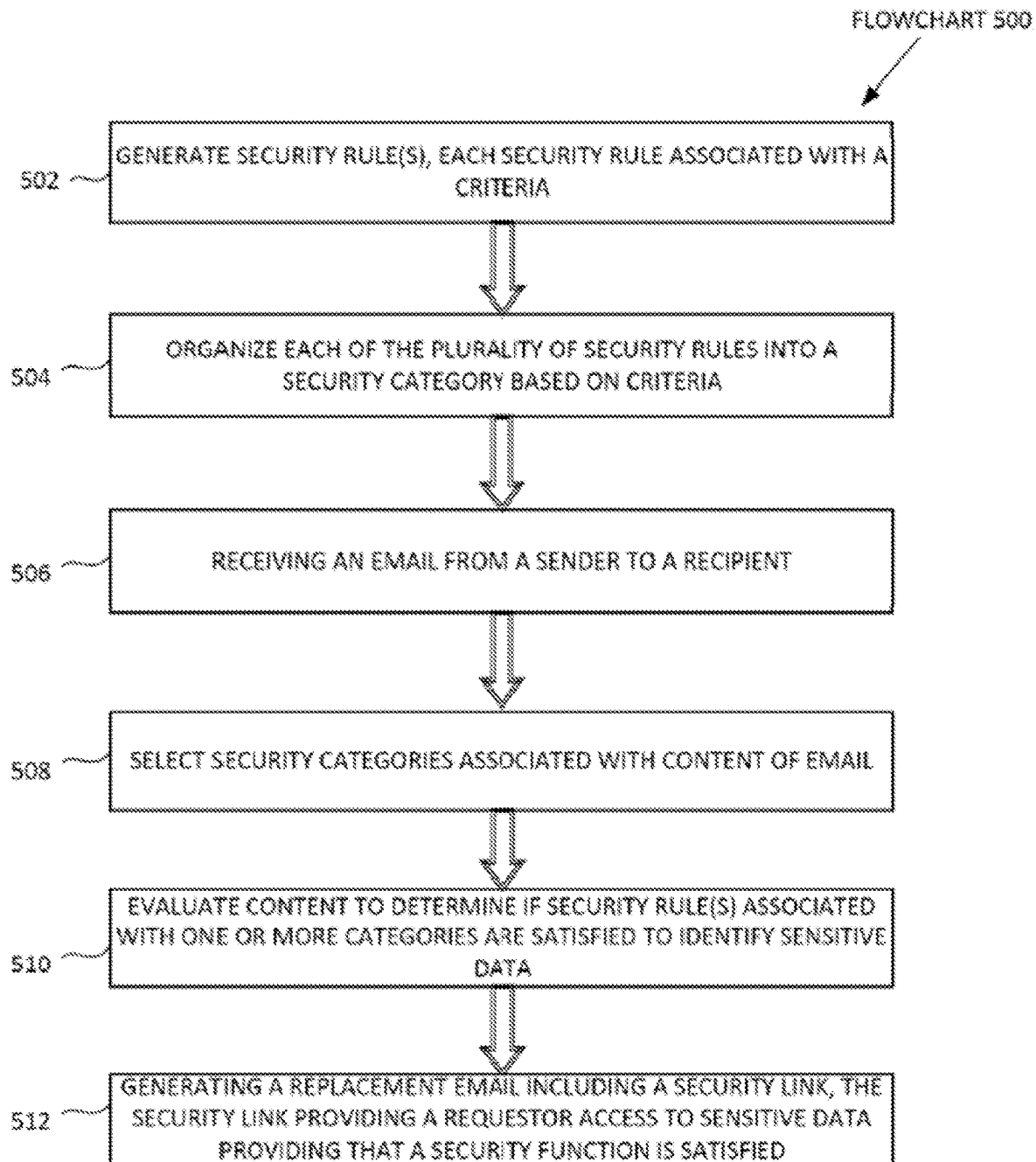


FIG. 5

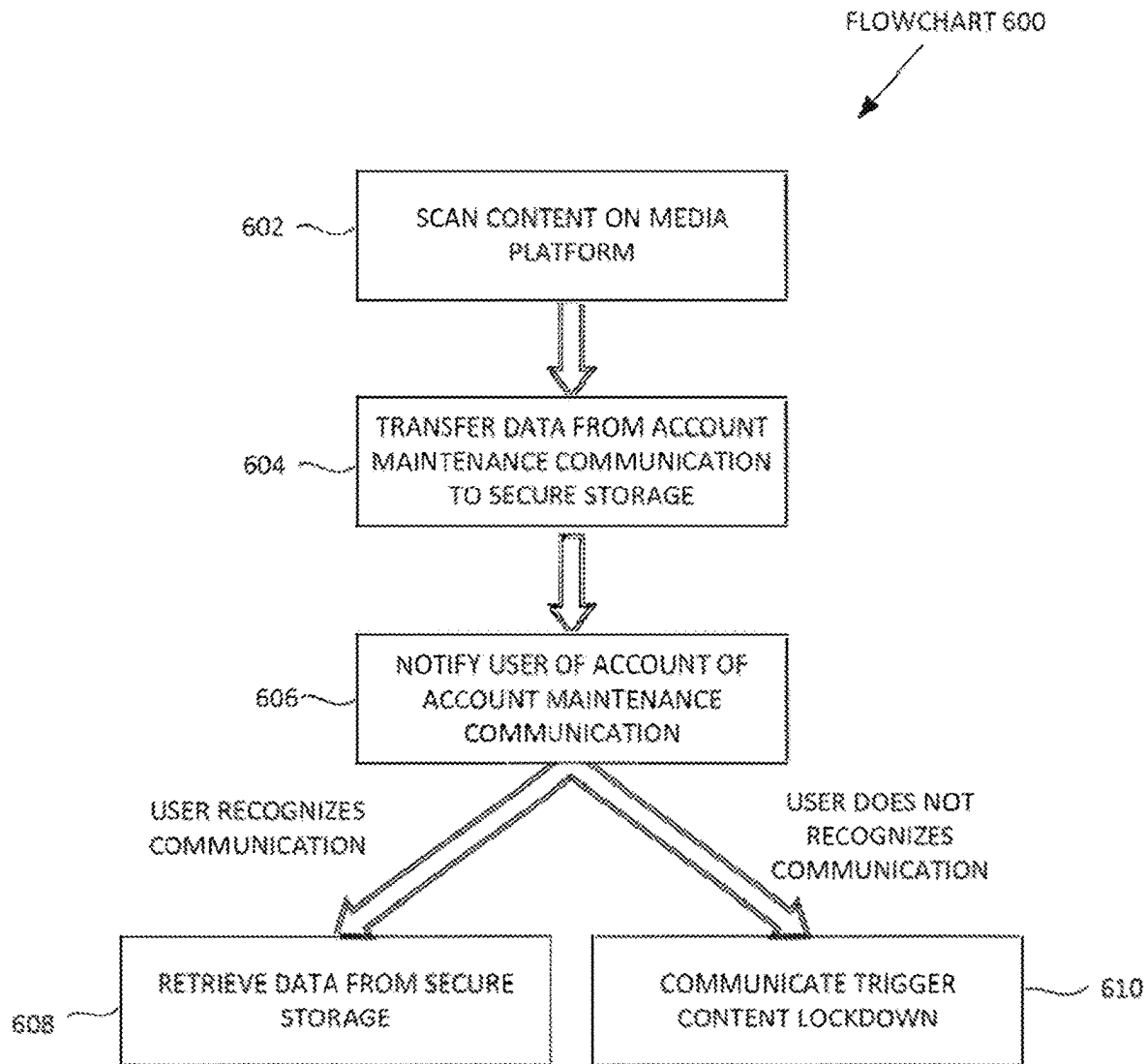


FIG. 6

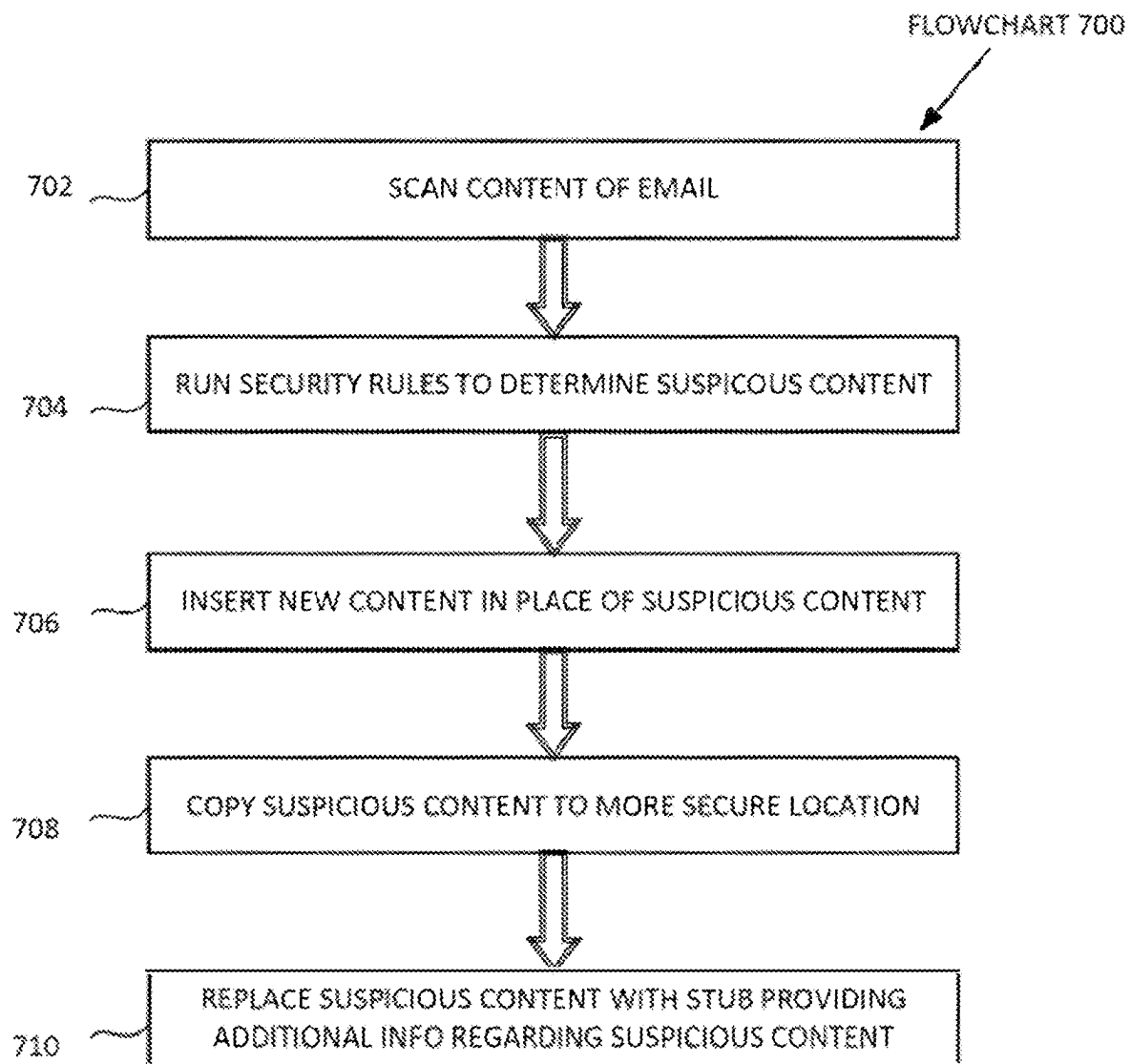


FIG. 7

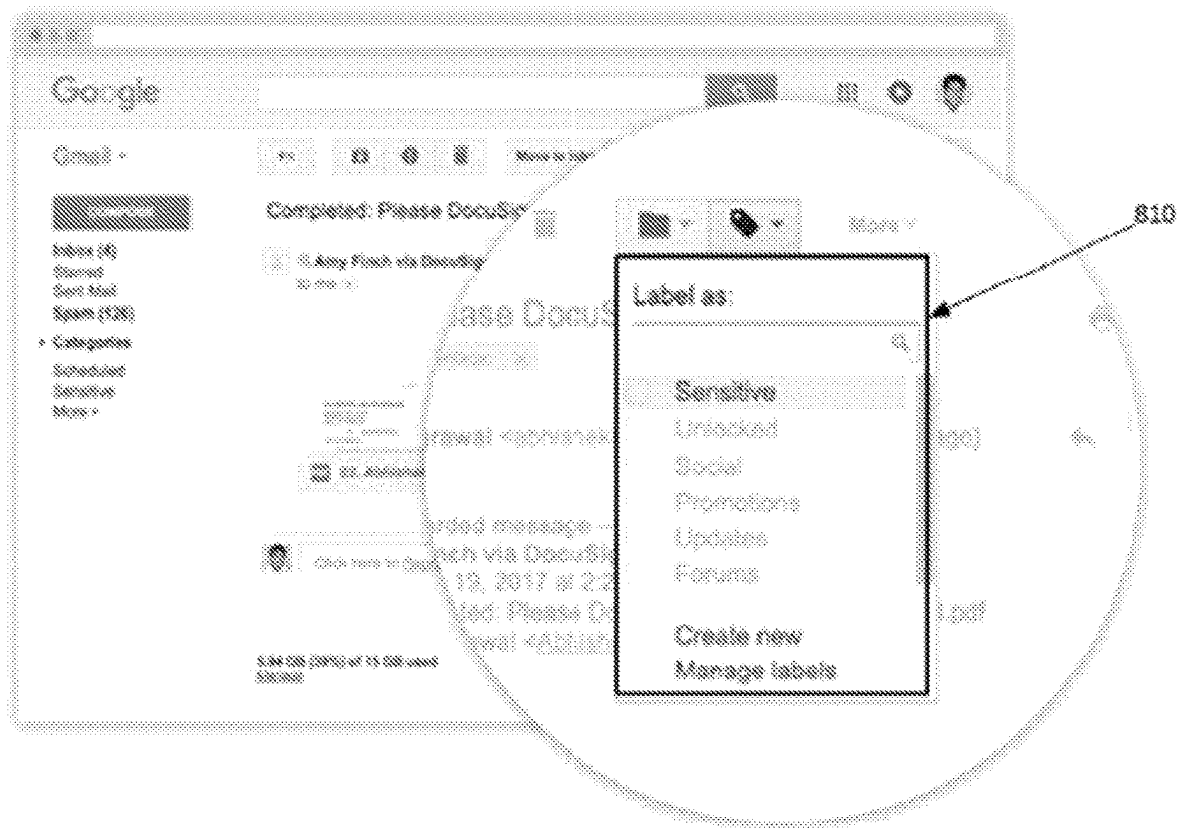


FIG. 8

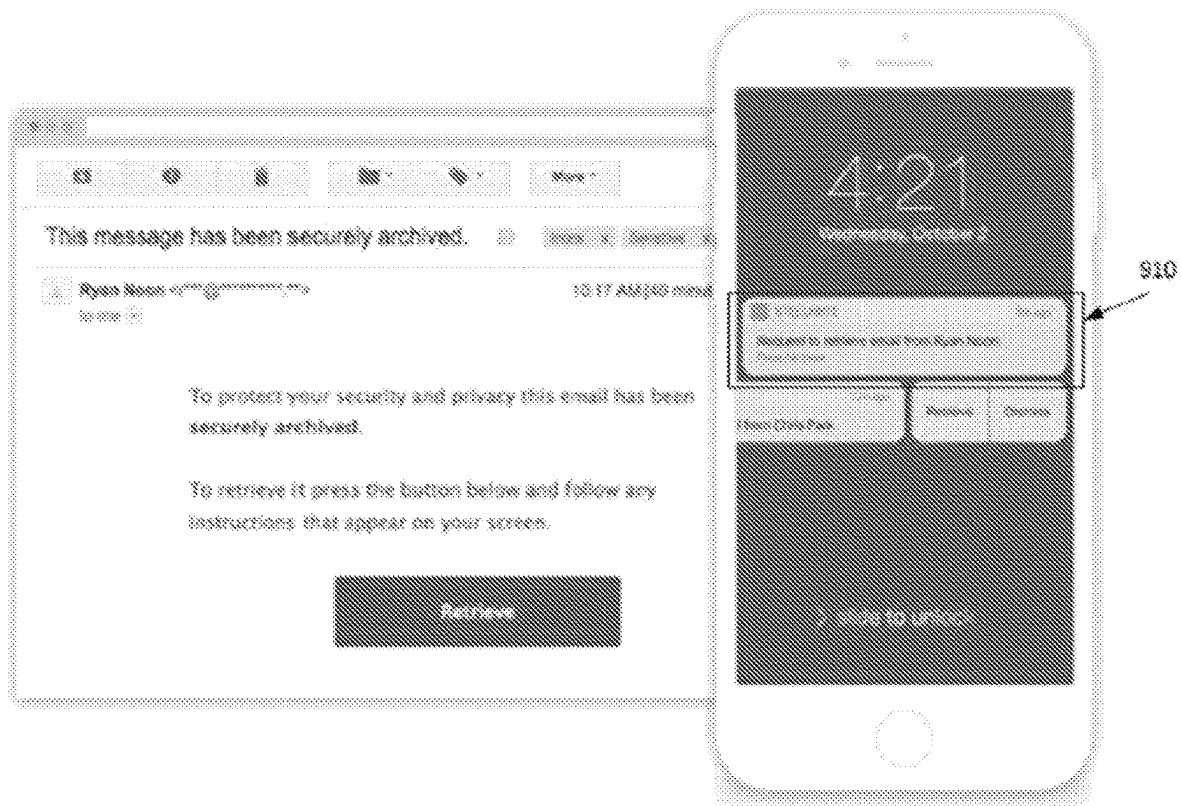


FIG. 9



FIG. 10

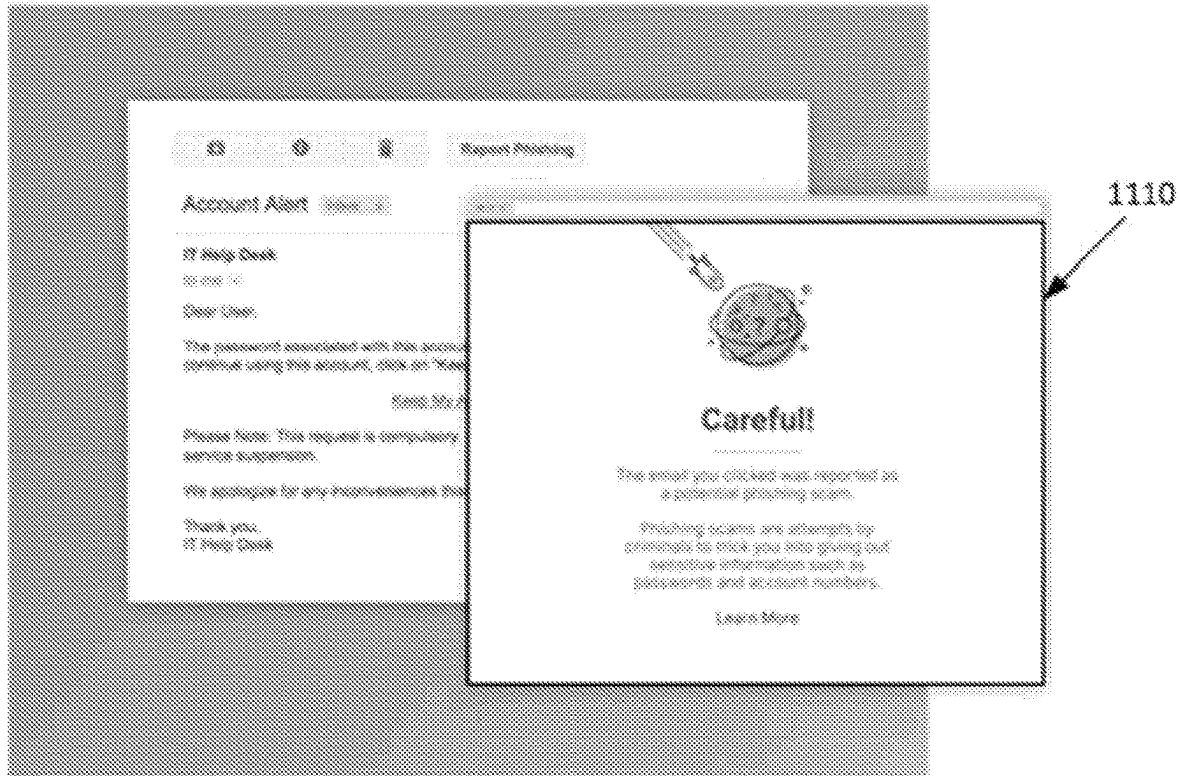


FIG. 11

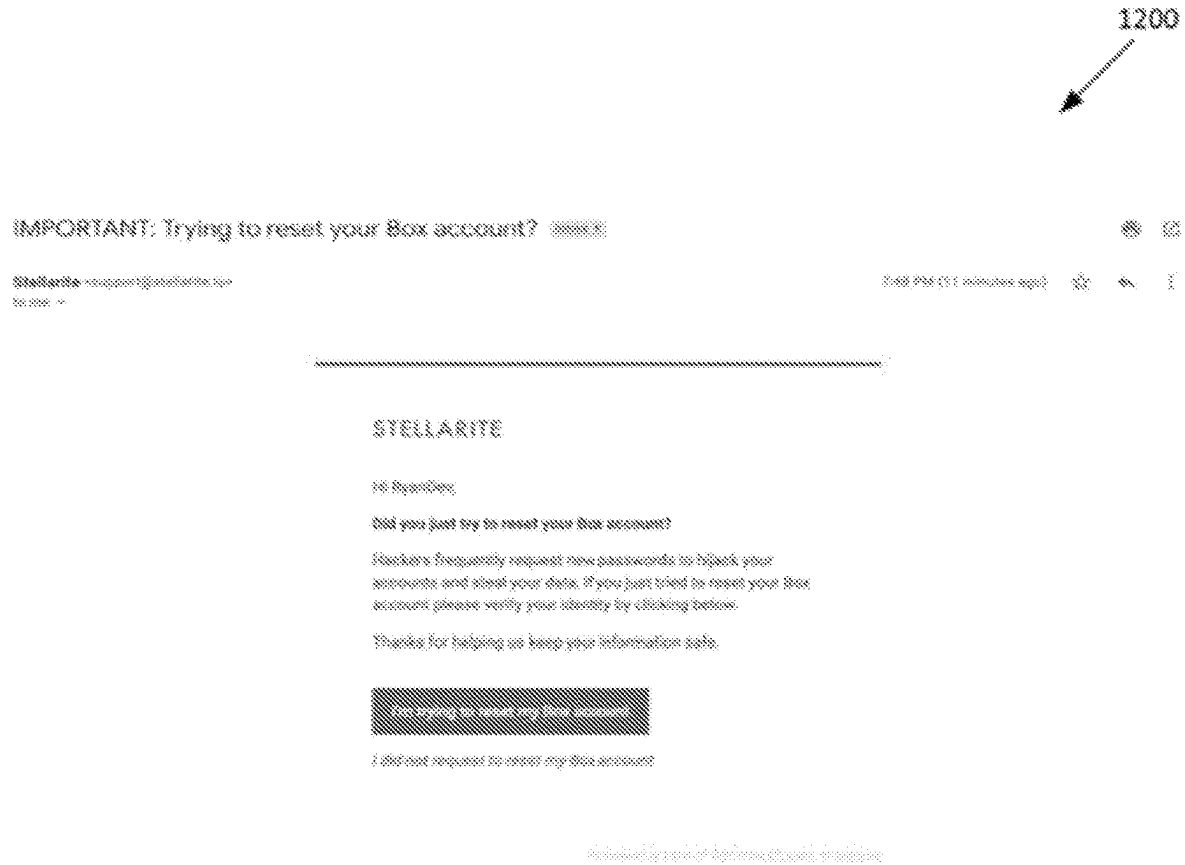


FIG. 12

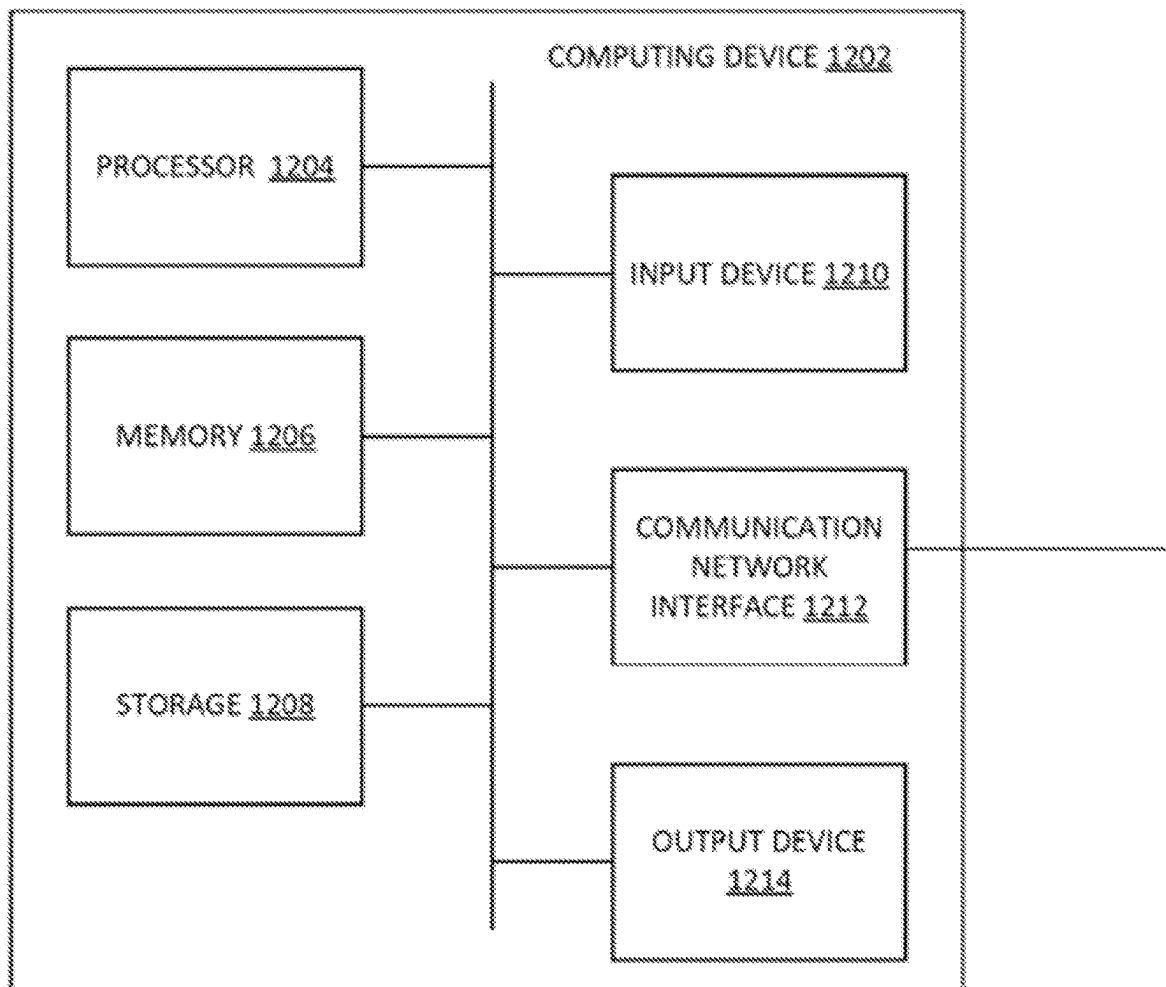


FIG. 13

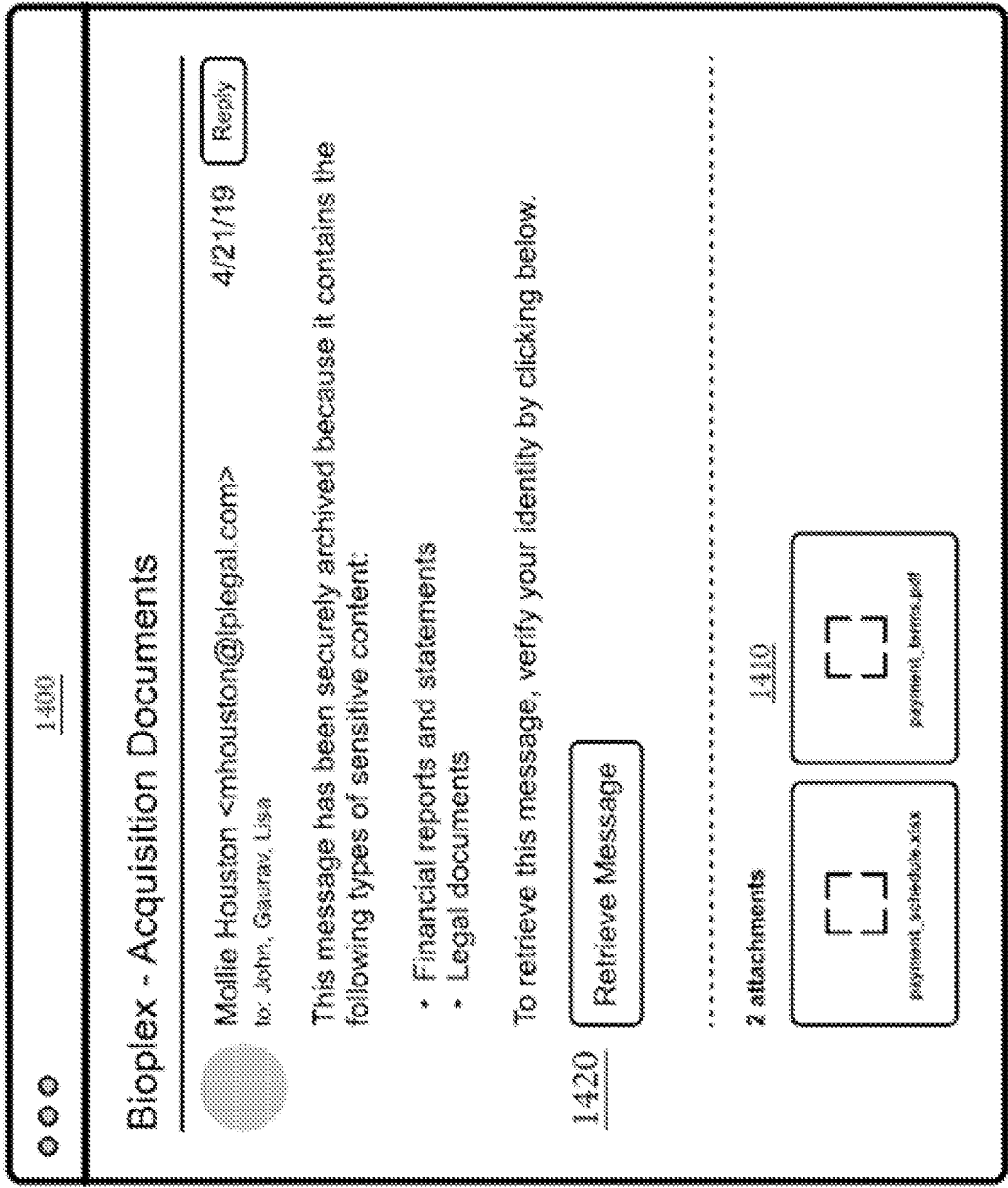


FIG. 14

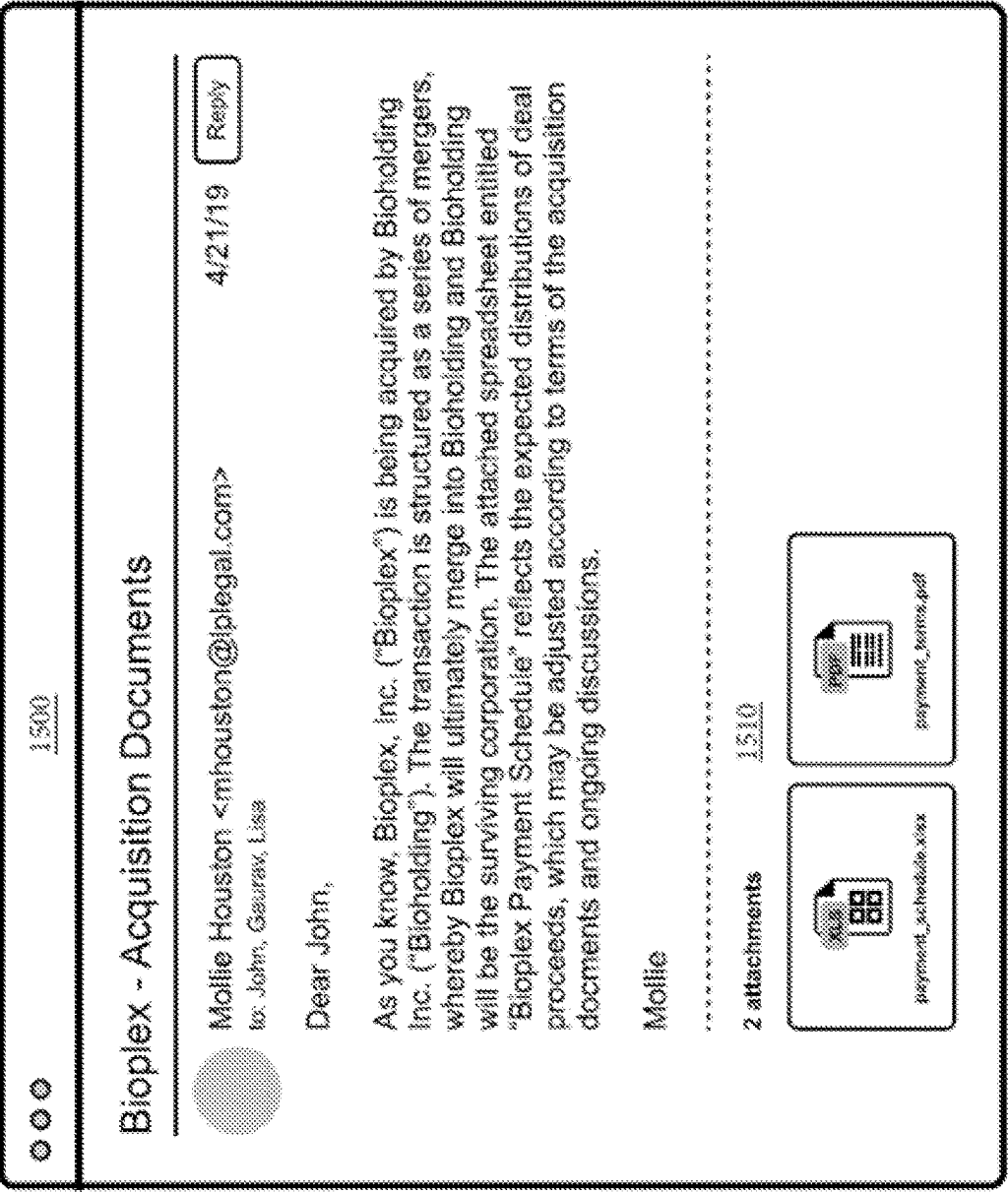


FIG. 15

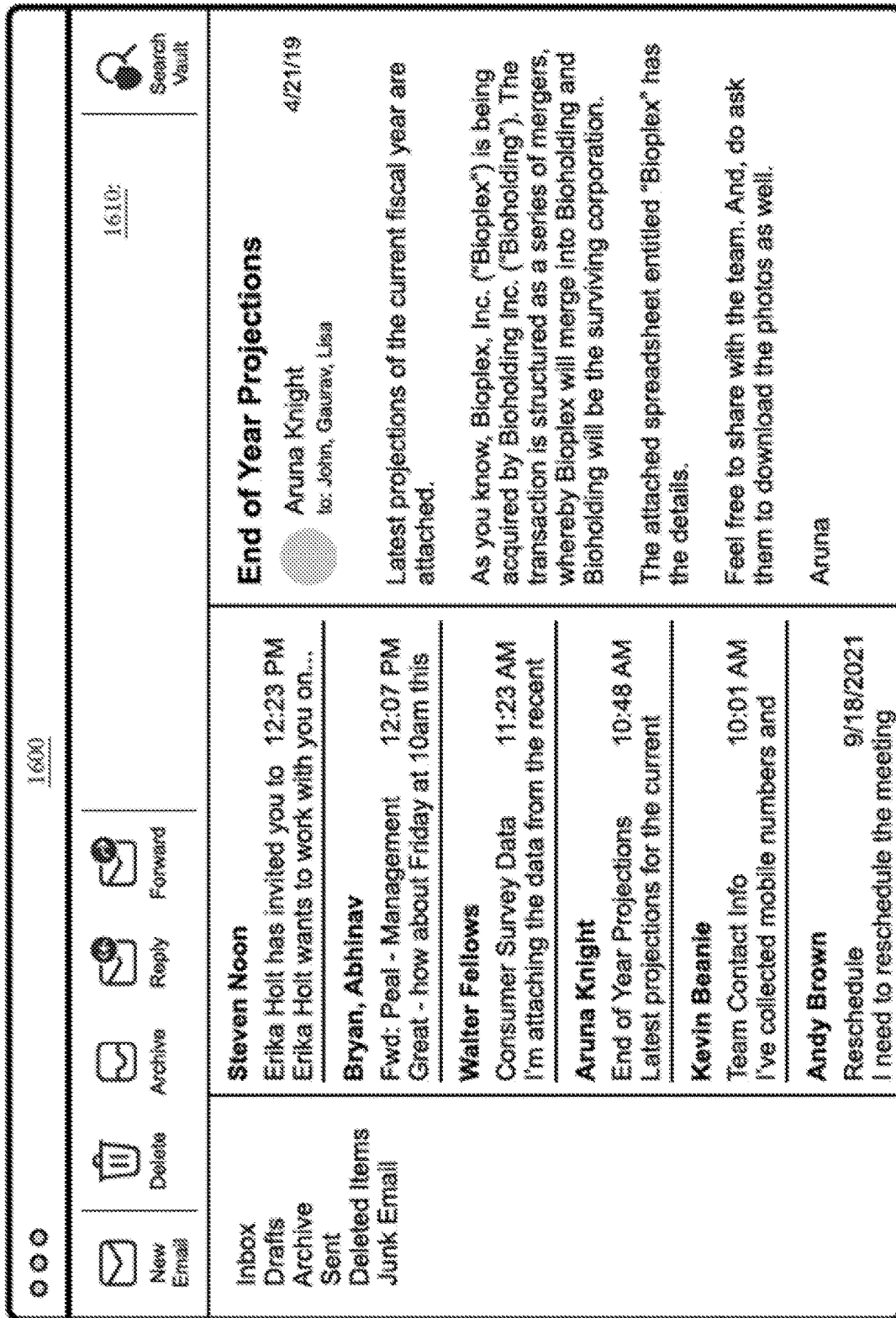


FIG. 16A

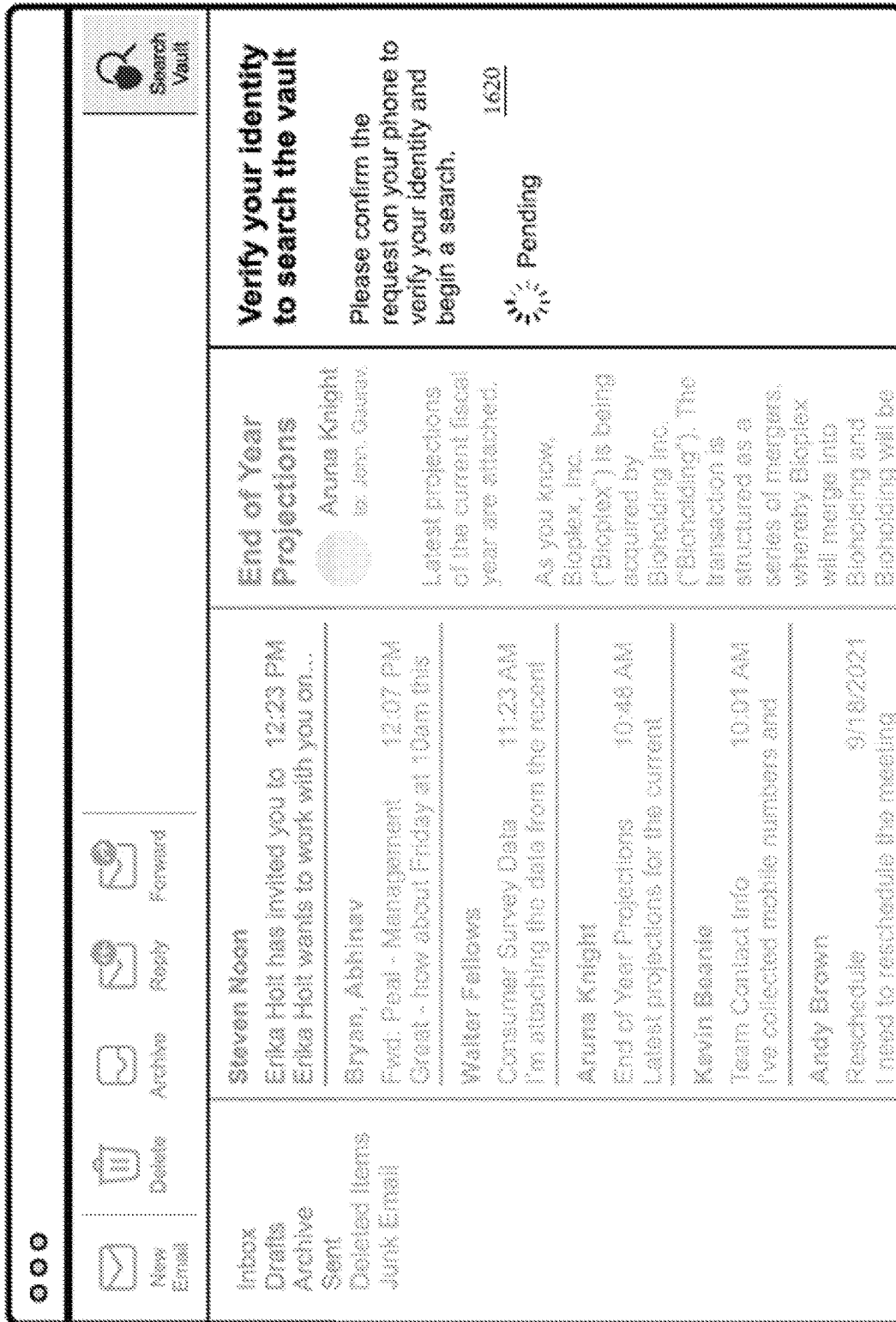


FIG. 16B

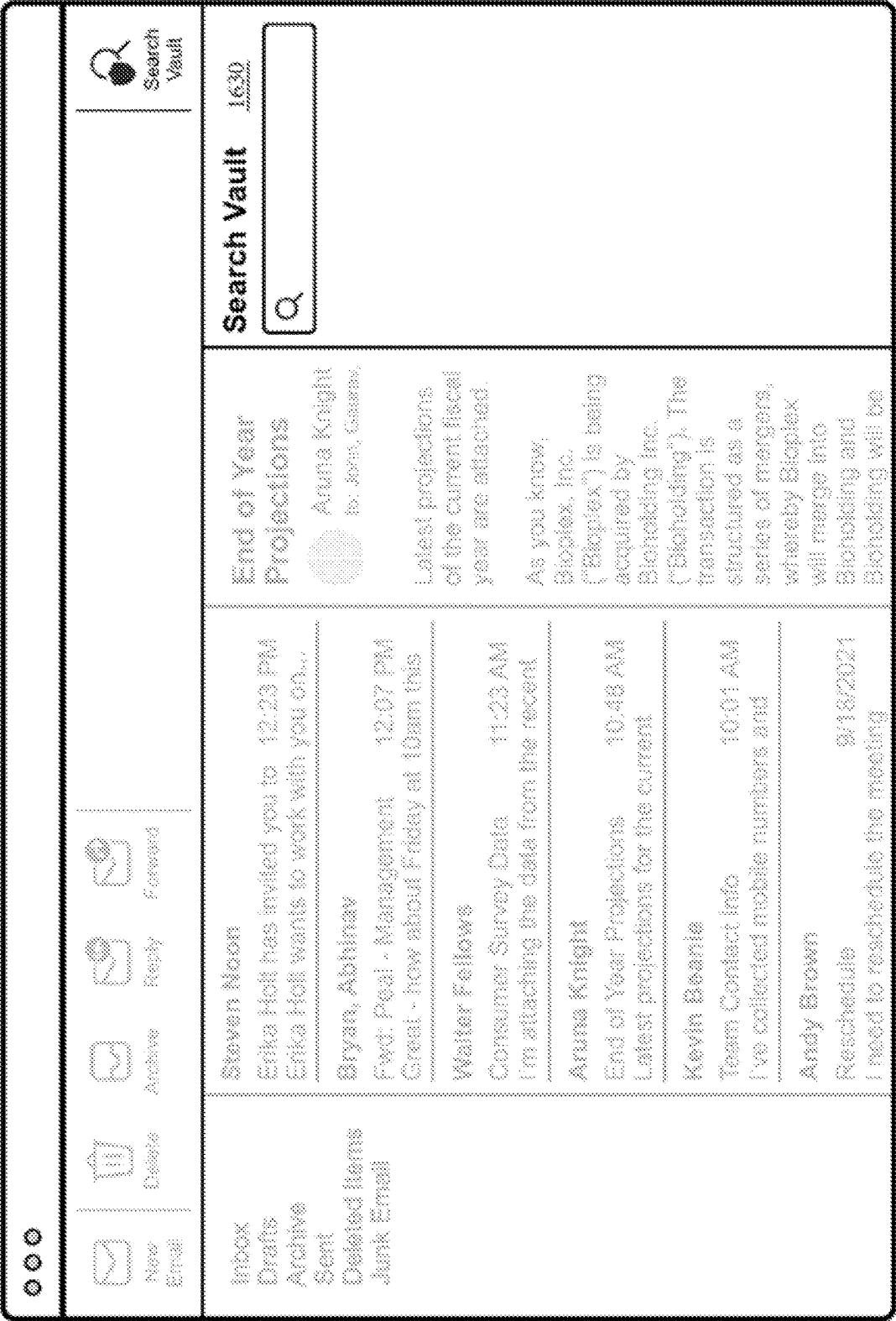
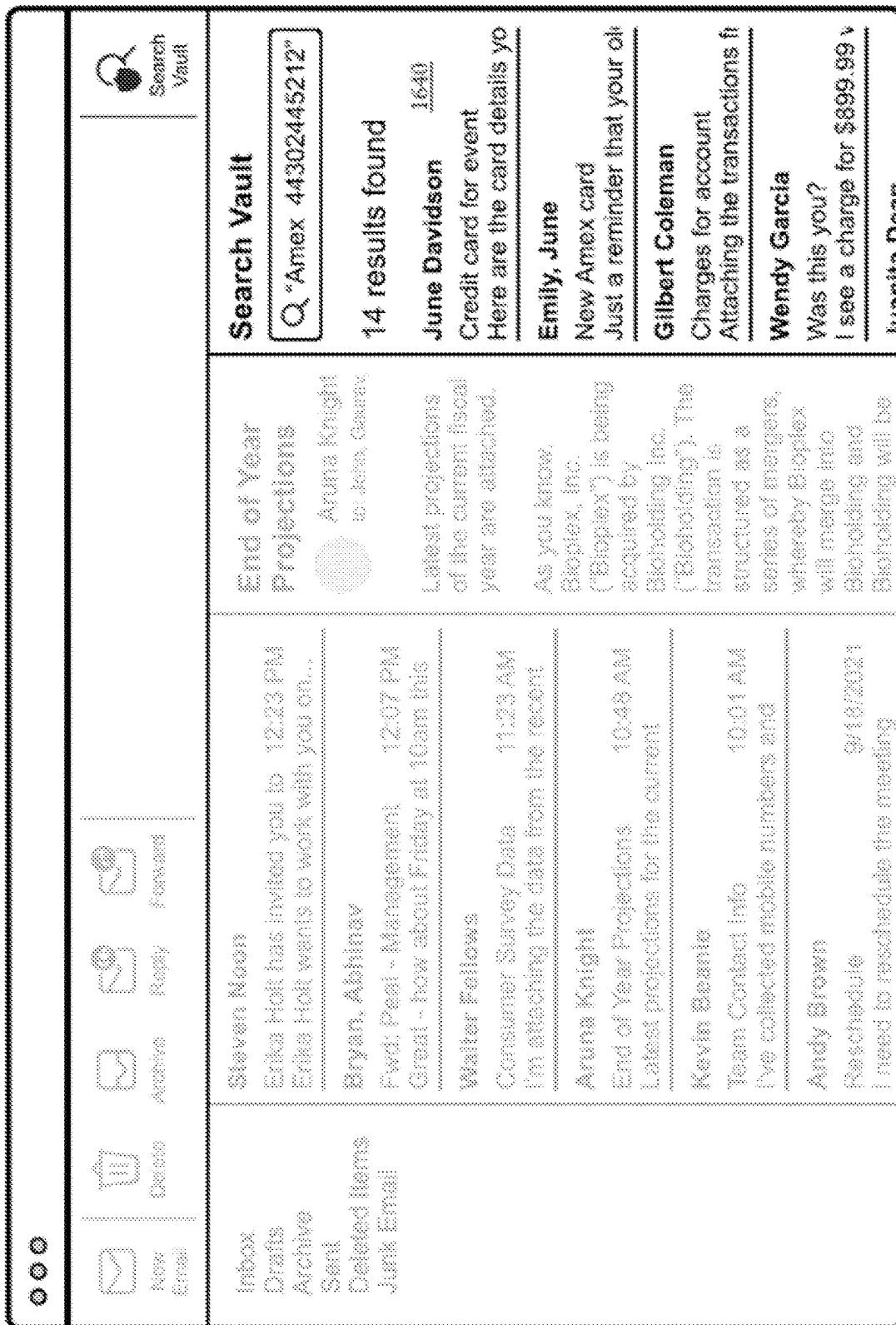


FIG. 16C



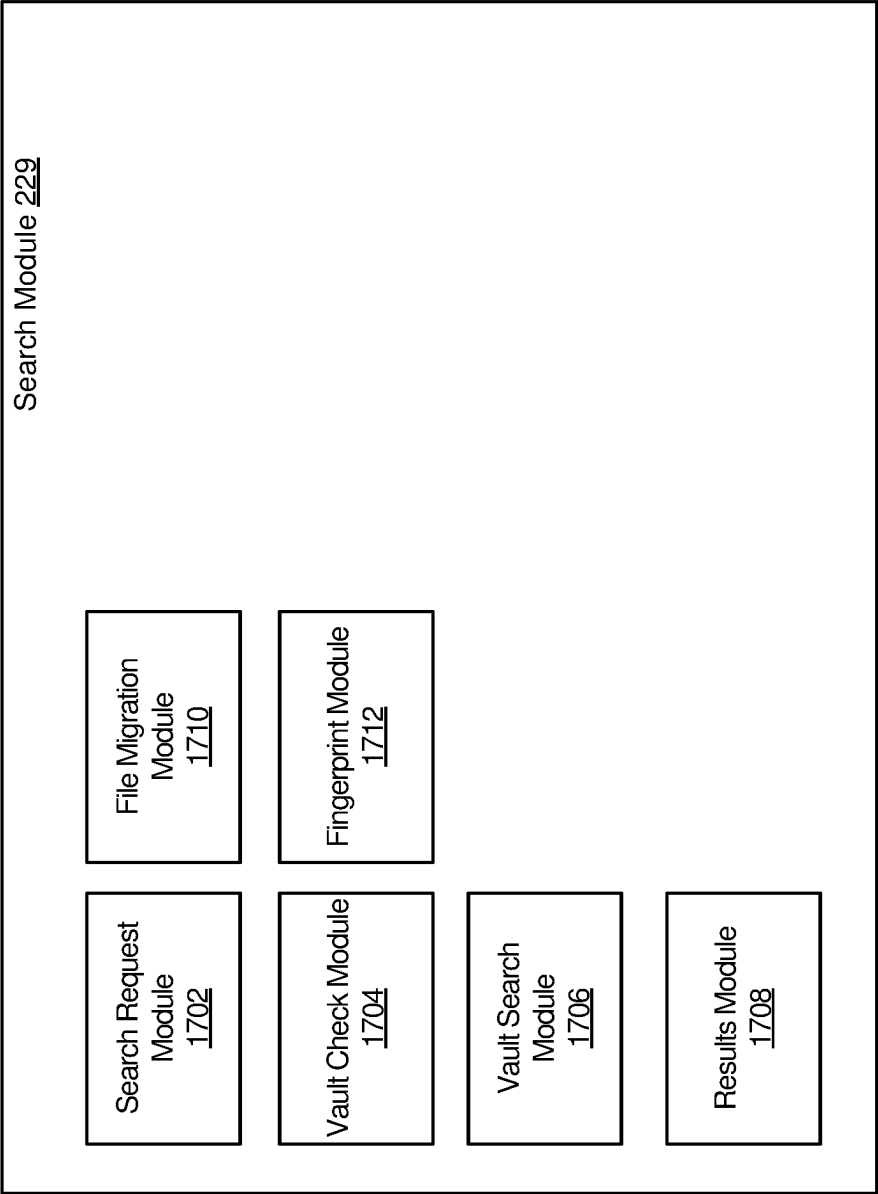


FIG. 17

1800:

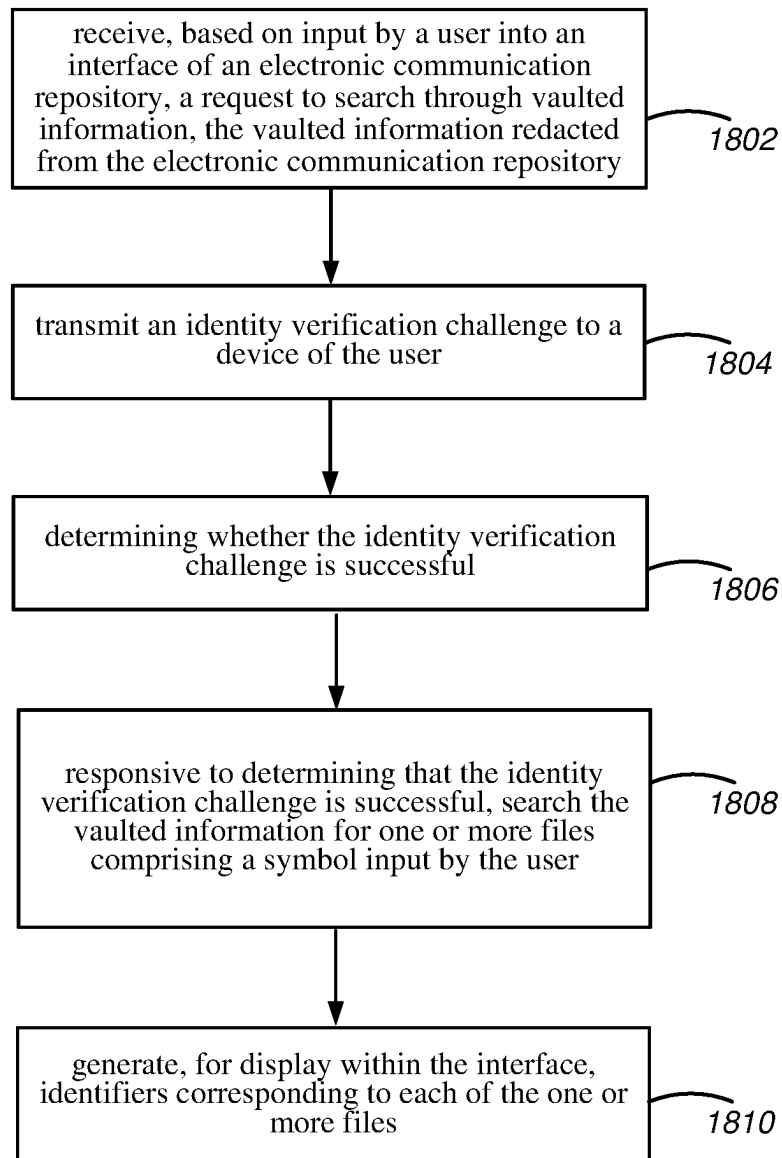


FIG. 18

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ONE-SHOT CHALLENGE TO SEARCH AND ACCESS UNREDACTED VAULTED ELECTRONIC COMMUNICATIONS

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/303,369, filed on Jan. 26, 2022. The content of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

Embodiments of the present invention related generally to protecting contents and accounts.

BACKGROUND

Current security systems are generally directed to blocking unauthorized users from accessing accounts. For example, there are numerous systems designed to challenge an attacker to block the attacker from logging into systems. Similarly, there are numerous anti-virus systems that scan email and files to find worms, viruses, or other forms of malware that may attack a system to either damage the system or to enable unauthorized users access. Systems may redact sensitive data from electronic communications and send the redacted electronic communications to an electronic repository, holding the original in vaulted security; however, search functionality to find messages having redactions will fail where search terms correspond to redacted information.

SUMMARY

Systems and methods are disclosed herein for searching information redacted from electronic communications. Electronic communications (e.g., emails, documents, files, and so on) may be redacted on the basis of rules. For example, the rules may be time-based (e.g., redact contents of an electronic communication X days after received), content based (e.g., redact contents of an electronic communication if a sensitive keyword is detected), etc. When a user attempts to access the redacted information, in some embodiments, the user may be required to do so on an individual electronic communication basis. That is, the user must find the electronic communication in question (which may be difficult if key terms for searching for the electronic communication are redacted), select an option for retrieving redacted information, and verify the user's identity to unlock what is redacted.

While such systems improve security in maintaining electronic communications dramatically over other solutions, the disclosure herein improves upon user experience in accessing those electronic communications securely. Using the disclosure herein, electronic communications may be redacted in search-friendly ways, such as removing attachments while leaving dummy attachments with the removed attachments' file names, thus enabling a user to search redacted electronic communications and have more success finding those electronic communications. Moreover, a user is enabled using the systems and methods disclosed herein to unlock access to searching and otherwise accessing the full contents of redacted electronic communications through passing an up-front challenge that unlocks search and/or access to "vaulted" (e.g., redacted) information.

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To this end, a tool is disclosed for searching information redacted from electronic communications. The tool receives, based on input by a user into an interface of an electronic communication repository, a request to search through vaulted information, the vaulted information redacted from the electronic communication repository. The tool transmits an identity verification challenge to a device of the user and determines whether the identity verification challenge is successful. Responsive to determining that the identity verification challenge is successful, the tool searches the vaulted information for one or more files comprising a symbol input by the user, and generates, for display within the interface, identifiers corresponding to each of the one or more files.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts a block diagram of an example system for protecting contents and account according to some embodiments.

FIG. 2 depicts a block diagram of an example of a secure content system according to some embodiments.

FIG. 3 depicts a flowchart of the process of securing contents according to some embodiments.

FIG. 4 depicts a flowchart of an example process of challenging a requester before retrieving sensitive data from the secure storage.

FIG. 5 depicts a flowchart of the process of securing contents according to various embodiments.

FIG. 6 depicts a flowchart of the process of account maintenance authentication according to some embodiments.

FIG. 7 depicts a flowchart of the process of suspicious content mitigation according to some embodiments.

FIG. 8 depicts an example email designating interface according to some embodiments.

FIG. 9 depicts an example notification interface according to some embodiments.

FIG. 10 depicts a communication interface according to some embodiments.

FIG. 11 depicts an example phishing interface according to some embodiments.

FIG. 12 depicts an example notification email according to some embodiments.

FIG. 13 depicts a block diagram of an example computing device according to some embodiments.

FIG. 14 depicts redacted stubs in an electronic communication and an option to retrieve what was redacted.

FIG. 15 depicts unredacted stubs in an unredacted electronic communication following identity verification.

FIGS. 16A-16D show a sequence of user interfaces involved in performing a search through multiple emails within a vault.

FIG. 17 depicts exemplary modules of a search module used to search vaulted information, in accordance with an embodiment.

FIG. 18 depicts an exemplary process for using the search module to search vaulted information, in accordance with an embodiment.

DETAILED DESCRIPTION

It is appreciated that a person or entity that obtains access to email, document, or files may still be unauthorized to receive the contents of the email, document, or files. In some examples, a malicious user may bypass security and log into a target system to obtain access to a document or intercept an email from a targeted user or system (e.g., through

redirection of email or spoofing). In yet another example, an employee may obtain email or documents from another part of their company even though they are not authorized to read or otherwise access the contents.

In at least some systems and methods described herein, a secure content system may intercept, receive, and/or scan email, document, and/or files. The system may identify information contained within the email, document, or files based on any number of rules (e.g., scanning for sensitive information based on sensitive data rules). The system may replace all or some of the information of the email, document, or files with any number of secure links. A user may obtain access to the email, document, or files, but otherwise may be unable to view the replaced information unless they engage the secure link. When the user interacts with the secure link, they may receive a challenge and/or security functions may be engaged to assist in determining if the user is authorized to receive the replaced content (i.e., the sensitive information that was replaced). If the challenge and/or security functions are satisfied, a secure content system may enable the recipient to access to the replaced information.

It will be appreciated that the system may include additional security functions. In some embodiments, the secure content system may limit access to the replaced information to a limited time duration, to a particular domain, to particular users/devices (e.g., MAC addresses), to particular network(s), to one or more particular users (e.g., users of a particular group), and/or the like.

FIG. 1 depicts a block diagram of an example system **100** capable of protecting contents and accounts according to some embodiments. In this example, system **100** comprises a sender user system **102-1** and a recipient user system **102-2** (individually, user system **102** collectively), communication network **104**, secure content system **106**, third-party cloud storage **108**, third-party email server **110**, enterprise client system **112**, and enterprise system **120**. The enterprise system **120** comprises enterprise network **122**, enterprise email system **124**, and enterprise data management system (DMS) **126**. User system **102**, secure content system **106**, enterprise client system **112** may each be or include any number of digital devices. A digital device is any device with a processor and memory. Digital devices are further discussed herein (e.g., see FIG. 12).

The user system **102** may be configured to facilitate communication between one or more users and other associated systems. In some embodiments, the user system **102** may be or include one or more mobile devices (e.g., smartphones, cell phones, smartwatches, table computer, or the like), desktop computers, laptop computers, and/or the like.

The communication network **104** may represent one or more computer networks (e.g., LAN, WAN, or the like). Communication network **104** may provide communication between any of user system **120**, secure content system **106**, and third-party cloud storage **108**, third-party email server **110**, and the enterprise system **120**. In some implementations, communication network **104** comprises computer devices, routers, cables, uses, and/or other network topologies. In some embodiments, communication network **104** may be wired and/or wireless. In various embodiments, communication network **104** may comprise the Internet, one or more networks that may be public, private, IP-based, non-IP based, and so forth.

Users may interact with user system **102** using, for example, a web browser or mobile application to communicate with other users, access content (such as email from third-party email server **110** or enterprise email system **124**), and/or access web pages on the enterprise client system **112**.

The users may also utilize user system **102** to interact with content, such as digital images, digital video, or the like stored in third-party cloud storage **108** or the enterprise data management system (DMS) **126**. The third-party cloud storage **108** may be or include Apple iCloud storage.

The secure content system **106** may scan contents associated with an authorized user and identify sensitive data by evaluating the contents based on one or more security/sensitivity rules. For example, the one or more security rules may define particular keywords and/or phrases as sensitive. The secure content system **106** may scan a document, file, or email for those keywords and/or phrases identified by the one or more security rules. For example, if those keywords and/or phrases are found during the evaluation of the email that a sending user wants to send to a recipient user, the identified sensitive data may be replaced and/or stored by the secure content system **106**. The term electronic communication, as used herein, may refer to any electronic file stored in association with a given user. For example, this may include e-mails, text messages, SMS messages, cloud files stored in association with a user account or local files, images, videos. This may also include attachments or embedded files within or outside of electronic communications (e.g., attachments, links, and so on). As a further example, electronic communications may include general chat messages (e.g., a slack message) and stored or archived copies thereof. Reference to "content" herein are interchangeably used with references to "electronic communications."

Any amount of the sensitive information may be stored by the secure content system **106**. In some embodiments, the secure content system **106** may store every paragraph containing sensitive information (e.g., containing any number of keywords or phrases known as sensitive based on the security rules, containing a number of keywords or phrases above a particular threshold defined by the security rules, and/or containing one or more particularly sensitive keywords or phrases relative to other keywords, phrases, or other content based on the security rules). In some embodiments, the secure content system **106** stores every sentence containing sensitive information. In various embodiments, the secure content system **106** determines a predetermined number of characters, words, phrases, sentences, paragraphs, sections, or the like in front and/or behind each keyword or phrase determined to be sensitive. The secure content system **106** may store the predetermined number of characters, words, phrases, sentences, paragraphs, sections, or the like in front and/or behind each keyword or phrase determined to be sensitive.

The secure content system **106** may generate a replacement electronic communication which contains a security link based on a security function to replace all or part of the sensitive data. The secure system **106** may replace the sensitive content (e.g., redact the sensitive content) with the security link (e.g., a stub). For example, the secure content system **106** may replace/redact every paragraph containing sensitive information (e.g., containing any number of keywords or phrases determined to be sensitive based on the security rules, containing a number of keywords or phrases above a particular threshold defined by the security rules, and/or containing one or more particularly sensitive keywords or phrases relative to other keywords, phrases, or other content based on the security rules). In some embodiments, the secure content system **106** may replace/redact every sentence containing sensitive information. As discussed herein, the secure content system **106** may determine a predetermined number of characters, words, phrases, sen-

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tences, paragraphs, sections, or the like in front and/or behind each keyword or phrase determined to be sensitive. The secure content system **106** may replace/redact the predetermined number of characters, words, phrases, sentences, paragraphs, sections, or the like in front and/or

As a result, the secure system **106** may replace any portion of an electronic communication (e.g., document, file, and/or email (or attachment attached to an email)) with a secure link that prevents a user accessing the particular document, file, and/or email from accessing the sensitive content without challenge. In one example, a user may receive an email containing sensitive information. Instead of opening the email and reading the sensitive information, the user may receive an email that includes a secure link that replaces all or some of the contents of the email (including the contents determined to be sensitive). In one example, there may be information (e.g., non-sensitive information) in the email and the sensitive content may be redacted and the secure link included in place of the redacted information. The recipient user may receive the replacement email and interact with the like to make a request to the secure content system **106** to access the sensitive data.

The function(s) triggered by interacting with the security link may challenge the recipient user and if the recipient user satisfies the security function, the sensitive data stored in the secure content system may be made available to the recipient user. For example, if the recipient user satisfies the security function(s), the secure system **106** may enable access to the original email, original attachment to the email (if any), and/or the sensitive content. For example, if the recipient user satisfies the security function, the secure system **106** may provide the sensitive content (e.g., redacted sensitive content) including one or more replaced/redacted paragraphs containing sensitive information (e.g., containing any number of keywords and/or phrases known as sensitive based on the security rules, containing a number of keywords and/or phrases above a particular threshold defined by the security rules, and/or containing one or more particularly sensitive keywords and/or phrases relative to other keywords, phrases, or other content based on the security rules), and/or the predetermined number of characters, words, phrases, sentences, paragraphs, sections, or the like in front and/or behind each keyword and/or phrase determined to be sensitive.

If the recipient user is not successful in satisfying the security function(s), the sensitive data may not be made available to the recipient user. In some embodiments, after a predetermined time period, the sensitive data associated with the security link may no longer be available to the recipient user and/or any other requester that engages the security link to access the sensitive information.

The predetermined time may be determined in any number of ways. For example, the predetermined time period may be configured by the sender of the email, default rules of an organization, rules of an enterprise, group membership of the sender, or the like. Alternately or in addition, the predetermined time period may be dependent on the status of the email (e.g., the predetermined period starts after the recipient opens the email or upon sending of the email). By securing sensitive data within the secure content system **106**, the sensitive data is secure from those that may obtain access to the email but are otherwise unauthorized to receive the contents of the email.

In some embodiments, the secure content system **106** stores one or more security rules. The security rules can be defined by the user system **102**, enterprise system **120** (e.g.,

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the sender's organization or a sub-organization of the enterprise of the enterprise system **120**), and/or pre-defined in the secure content system **106**. Each of the one or more security rules may be organized into security categories depending on criteria associated with each of the security rules. Security categories may organize security rules according to industry (e.g., healthcare, military, banking, or the like), organizational structure (e.g., temporary employees, external contractors, interns, executive, departments, or the like), sensitivity of information (e.g., personal information, client information, accounting information, trade secrets, technical information), remote vs. local recipients, recipients in other organizations or sub-organizations within the enterprise, and/or the like.

In various embodiments, different enterprise systems using the secure content system **106** may organize their security rules in their own set of security categories (e.g., security buckets). In some embodiments, different enterprise systems or user systems may store sensitive data in a respective partition of the secure content system **106**, so that, for example, if an unauthorized user was able to gain access to the user system **102** and the sensitive data defined by the user system **102**, sensitive data defined by the enterprise system **120** may be secure from the unauthorized user.

It will be appreciated that the secure content system **106** may receive or intercept an email, file, or document, redact sensitive information, store the sensitive information, generate a new email, file, or document with a secure link in place of all or some of the sensitive information (and potentially information that is not sensitive), and send the email, file, or document to the recipient.

In some embodiments, an originating sender may send an electronic communication (e.g., e-mail, SMS message, group chat message, and so on) using the enterprise email system **124** to a first plurality of recipients. The remainder of this example refers to e-mail for convenience, but any other electronic communication format may be used. Continuing with the email example, if one of the first plurality of recipients designates the email or any of its contents or attachments (e.g., content of an attachment to the email) as sensitive, some or all parts of the email may be transferred from the enterprise email system **124** to the secure content system **106**. Additionally, the email may be designated as sensitive for the remaining first plurality of recipients as well as the originating sender. If one of the first plurality of recipients, who may or may not be the recipient who designated the email as sensitive, forwards the email to a second plurality of recipients, the sensitive email designation may propagate to the second plurality recipients, and any subsequent recipients of the email. As a result, the secure content system **106** may intercept the email, scan the email and attachments for sensitive information, replace the sensitive information with a secure link, and provide the email including the secure link to the next recipients.

In some embodiments, a user of user system **102** may authorize the secure content system **106** to access the user's content such as an email account on the third-party email server **110** or a cloud storage account on the third-party cloud storage **108**. Again, e-mail is merely exemplary, and any storage account may be referenced in place of email (e.g., SMS repository, group chat repository, cloud file storage account, and so on). Following with the e-mail example, in various embodiments, the secure content system **106** may configure the user's account or software to enable directing sent email from the user's email account to scan for secure content. In various embodiments, the secure content system **106** may associate an enterprise and/or security

policies with the user and the user's email. In this example, the secure content system **106** may be configured to receive email sent by the user from the third-party cloud storage **108** and scan the email and/or attachments to the email for sensitive information based on the security policies. The third-party cloud storage **108** may replace all or some of the sensitive information with a security link (e.g., in the email and/or attachments to the email), before sending the email (or a replacement email as described herein) to the recipient.

In some embodiments, the secure content system **106** may monitor the user's email on the third-party email server for malicious attacks such as email including account maintenance communication requests such as a password reset request email. The secure content system **106** may store sensitive data from the account maintenance communication (e.g., a Uniform Resource Locator (URL) of the internet address to reset the password) or other information (e.g., such as account number which may be included as a part of the account maintenance communication to the secure content system **106**). The secure content system **106** may notify the user using an alternate method of communication, such as a Short Message Service (SMS) or Multimedia Messaging Service (MMS) to contact the user and determine if the user initiated the account maintenance communication. If the secure content system **106** does not receive an indication that the user initiated the account maintenance, the secure content system **106** may lock the account or redact content requesting information that may be as a result of a phishing attack, spear phishing attack, or social engineering to prevent unauthorized access to the user's electronic accounts.

The third-party cloud storage **108** may provide remote data storage for digital content for one or more user systems **102** and/or the enterprise system **120**. Cloud storage is becoming more popular and more users are using remote data servers such as the third-party cloud storage **108** to store digital documents and software or applications. For example, many users today store their personal email and contact information, and even pictures, videos, and music archives on remote servers, and access that data using third-party applications that are provided through and controlled by a web-browser.

The third-party email server **110** may provide storage for email, such as Google Mail™, Hotmail™, and the like. The third-party email server **110** may distribute and synchronizes across multiple mailboxes and over wired and wireless networks.

The enterprise client system **112** may provide access to any users on system **100** of web pages hosted by the enterprise client system **112**. In another example, the enterprise client system **112** may include remote users (e.g., recipients or senders) that receive or provide documents, files, or email with other users and/or devices (e.g., with one or more devices of the enterprise system **120**, the sender user system **102-1**, the recipient user system **102-2**, and/or the like).

The enterprise system **120** may be any digital devices that provide resources, services, documents, files, email, media, and/or the like to one or more enterprises or organizations. In one example, the enterprise system **120** includes internal servers, networks, network devices, workstations, storage devices, application servers, document management servers, email systems, and/or the like. While the enterprise system **120** depicts a single network with two devices, it will be appreciated that the enterprise system **120** may include any number of networks with any number of devices. Further, it will be appreciated that the enterprise system **120** may include cloud resources such as Amazon S3 services, and/or

the like that are used by the enterprise for enterprise-related businesses and/or functions. In the example depicted in FIG. **1**, the enterprise system **120** may comprise the enterprise network **122**, the enterprise email system **124**, and the enterprise DMS **126**.

In some embodiments, enterprise network **122** represents one or more computer networks (e.g., LAN, WAN, or the like). Enterprise network **122** may provide communication between any of the enterprise email systems, the enterprise DMS, any user of the enterprise system **120** including user **102-1** and other components of system **100**. In some implementations, enterprise network **122** comprises computer devices, routers, cables, uses, and/or other network topologies. In some embodiments, enterprise network **122** may be wired and/or wireless. In various embodiments, communication network **104** may comprise the Internet, one or more networks that may be public, private, IP-based, non-IP based, and so forth. In various embodiments, enterprise network **122** includes a firewall or other forms of hardware or software that helps screen out hackers, viruses and worms trying to reach electronic components connected to the enterprise network **122**.

The enterprise email system **124** may be any systems and/or services for enabling creation, transmission, receipt, storage, reading, accessing, viewing, configuring settings, and/or the like of electronic communications (e.g., sent internally within the enterprise system **120** or externally outside the enterprise system **120**).

The enterprise document management system (DMS) **126** may be used to hold and manage files (e.g., documents, images, audio, data, and/or the like) of the enterprise system **120**. The enterprise DMS **126** may include information regarding employee records, client records, and other documents related to enterprise system **120**.

It will be appreciated that content (e.g., sensitive and non-sensitive information) may include any kind of digital content or media, including, but not limited to pictures, text, video, sound, graphics, icons, interactive programming, or any combination of the above.

While the secure content system **106** is depicted as being in communication with the enterprise system **120**, it will be appreciated that any number of different enterprises with different systems, users, email servers, and the like may utilize the same or any number of secure content systems **106**. Similarly, there may be any number of third-party cloud storages **108**, third-party email servers **110**, and the like. In some embodiments, the security content system **106** is part of the enterprise system **120**, and is not shared by different enterprises, with security rules and policies and secure storage all within the control of the enterprise system **120**.

FIG. **2** depicts a block diagram of the secure content system **106**, from FIG. **1**, according to some embodiments. The secure content system **106** includes communication engine **202**, controller engine **204**, content scanning engine **206**, security policy engine **208**, security rule datastore **210**, sensitive data engine **212**, secure storage **214**, security link generator **216**, replacement engine **218**, security engine **220**, sensitive data provider engine **222**, invitation engine **224**, content datastore **226**, notification engine **228**, and search module **229**.

The communication engine **202** may be configured to send and receive requests or data between secure content system **106** and any other component of system **100**. In various embodiments, any number of devices of the enterprise system **120** may be configured to provide electronic communications (e.g., documents, files, and/or email (e.g., any item or object containing text)) to the secure content

system **106**. In various embodiments, the enterprise email system **124** may be configured to direct email and/or other messages (e.g., SMS messages and the like) to the secure content system **106** prior to the message being directed to the intended recipient. Alternately, the secure content system **106** may otherwise intercept the messages. Similarly, in various embodiments, the enterprise email system **124** may be configured to direct documents and/or other files (e.g., SMS messages and the like) to the secure content system **106** prior to storing to, while being stored at, or prior to access from the Enterprise DMS **126** or any other storage. Alternately, the secure content system **106** may otherwise intercept the documents and/or other files using the communication engine **202**.

The controller engine **204** may be configured to manage one or more of the various engines shown in FIG. 2. For example, the controller engine **204** may be configured to scan the email account of a user subscribed to secure content system **106** by facilitating the content scanning engine **206** to evaluate the contents of one or more email based on security rules to identify sensitive or suspicious data. In some embodiments, the controller engine **204** may be configured to respond to a security link generator request by facilitating the security link generator **216** to create one or more security links associated with sensitive or suspicious data.

The content scanning engine **206** may be configured to scan contents of an electronic communication (e.g., email, email attachment, document, file, or other item containing content). The content scanning engine **206** may scan contents (e.g., contained text or image(s) of contained text) for sensitive information based on one or more security policies (e.g., stored and/or created by the security policy engine **208**).

The content scanning engine **206** may be configured to evaluate content such as email and/or email attachments from the third-party server **110** or enterprise email system **124** (e.g., subscribed to the secure content system **106**). The content may be contained within the email and/or email attachments. The content may include textual components or images of such textual components.

In some embodiments, the content scanning engine **206** may perform textual recognition processes on the contents of the email. In some embodiments, the content includes digital images or digital video. In some embodiments, the content scanning engine **206** may perform image recognition or pattern recognition processes on email, email attachments, files, and/or documents. The content scanning engine **206** may be configured to evaluate the contents of the email, email attachments, files, and/or documents based on security rules to identify sensitive data or suspicious data.

In various embodiments, the content scanning engine **206** may perform optical character recognition on any number of electronic communications, electronic communication attachments, files, and/or documents including, for example, Adobe Acrobat files, images, and/or the like. It will be appreciated that the content scanning engine **206** may retrieve or receive any number of electronic communications (e.g., email, email attachments, files, and/or documents). For example, the content scanning engine **206** may identify the type or file attached to an email in order to determine what kind of processing may be required. The content scanning engine **206** may perform different processing on different file email attachment types. For example, the content scanning engine **206** may perform optical character recognition on image files or Adobe Acrobat files prior to scanning for sensitive information. In another example, the

content scanning engine **206** may receive a Microsoft Word or other text file, and immediately scan for sensitive information without performing other operations. The content scanning engine **206** may perform any number of operations based on any type of electronic communication (e.g., email, email attachments, files, and/or documents).

The security policy engine **208** may be configured to create and/or retrieve one or more security policies. Each security policy may contain any number of security rules, where each security rule may indicate what information in an electronic communication (e.g., document, file, or email) should be considered as sensitive. In one example, a security policy may contain any number of keywords or phrases that are associated with being sensitive. Any number of keywords or phrases may be associated as being sensitive. For example, keywords such as, but not limited to, accounting, technical, secret, intellectual property, proprietary, confidential, privileged, and the like may be considered to be indications of sensitivity by one or more security policy. It will be appreciated that keywords or phrases that are specific to the organization or sub-organization of an enterprise (e.g., groups of employees associated by department, function, seniority, subsidiary, relationship, and/or the like) may associated as being sensitive information. For example, keywords associated with specific technical, finance, personal, employee, software, or the like that is specific to the enterprise may be considered confidential by the enterprise. Similarly, keywords regarding executives, departments, domains, groups, projects, and or the like may also be considered to be sensitive. While keywords are described, it will be appreciated that phrases parts of words sentences and/or the like could be identified by one or more policies as being sensitive information. As described herein, keywords may refer to specific words, parts of words, phrases, paragraphs, sentences and/or any combination of words.

There may be any number of security policies. Each group within an enterprise may have one or more security policies and the keywords and phrases of each security policy of the group and/or the enterprise may be separate (e.g., exclusive) or contain any amount of similarity or overlap.

In various embodiments, a security policy may indicate how information is to be redacted from an email, document, or file. For example, the security policy may indicate that any paragraph or sentence containing one or more keywords or phrases determined to be sensitive should be redacted from the email, document, or file. In various embodiments, an employee, administrator, email sender, document creator, or the like, may define the redaction/replacement policy. In one example, an administrator may define a predetermined number of sentences, phrases, words, characters, and/or the like to redact or replace when a sensitive keyword or phrase is detected. In another example, an administrator may define a predetermined number of sentences, phrases, words, characters, and/or the like to redact or replace when a predetermined number of sensitive keywords and/or phrases are detected (and/or are when the predetermined number of sensitive keywords and/or phrases are detected within a predetermined number of paragraphs, sentences, phrases, words, or characters of each other). There may be different policies depending on the number and proximity of sensitive words and phrases to each other. For example, the administrator may include within the security policy that if there are three or more sensitive keywords and/or phrases within a paragraph, the paragraph should be replaced with a security link.

Alternately, the administrator may include within the security policy that each sentence containing any number of

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sensitive keywords and/or phrases should be redacted. In this example, if a user or recipient wishes to access the sentences containing the sensitive keywords and/or phrases, the user or recipient will have to overcome the security challenge(s) associated with the security link. The document, file, or email may include a single security link with any number of security functions to challenge the user (e.g., a password requirement) before the user may obtain access to all or some of the sentences containing the sensitive keywords and/or phrases.

The security policy engine **208** may be configured to select one or more security categories to apply to the content scanned by the content scanning engine **206**. For example, depending on the user system **102** or enterprise system **120** that created, is to receive, that requested, that modified, that sent, and/or is otherwise associated with the email, attachment, document, or file, the security policy engine **208** may select different security categories to apply to the content. For example, a first security category which is applicable to the user system **102** may not be applicable to the enterprise system **120**.

For privacy and security reasons, a particular user system or enterprise system may select from one or more security categories that are exclusive to the particular user system or enterprise system. Security categories may organize security rules according to industry (such as healthcare, military, banking), organizational culture (such as temporary employees, external contractors, interns, executive), domain, group membership, department membership, listserv, etc. In some embodiments, security rules may be further organized in levels of security, such as various levels of confidentiality, trade secret, and for executive eyes only. Once a security category has been selected, any number of security rules associated with the selected security category stored in the security rule datastore **210** may be used to evaluate the contents of the email.

Security rules and/or policies may be defined by a user, an enterprise, and/or may be pre-defined by the secure content system **106**. Security rules may be organized into security categories depending on the criteria associated with each of the one or more security rules. Security categories may organize security rules according to industry (such as healthcare, military, banking), organizational culture (such as temporary employees, external contractors, interns, executive), domain, group membership, department membership, listserv, etc. In various embodiments, different enterprise systems using the secure content system **106** may organize their security rules in their respective security policies.

In some embodiments, an employee or other user may identify an electronic communication (e.g., email), attachment, keywords within the electronic communication and/or attachment, phrases within the email and/or attachment, or the like as containing sensitive information. For example (using email as a representative example that can be generalized to all electronic communication types), a sender of an email may identify the contents of an email or the email itself as containing sensitive information through the use of a menu email program or an application associated with the email application. The sensitive data engine **212** may receive these indications of data sensitivity and further update security rules and/or policies to default such information as being sensitive for future scans (e.g., adding keywords and/or phrases as being sensitive). As a result, if a second employee includes similar content previously marked as confidential or sensitive by a different employee, the secure content system **106** may identify such information as being

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sensitive even if the second employee did not mark the email or the content as being specifically sensitive.

Similarly, if the first employee identifies information as not being sensitive, then the sensitive data engine **212** may update the security rules and/or security policies to indicate that such information is not sensitive. It will be appreciated that the sensitive data engine may make a statistical determination of keywords and/or phrases in order to identify content as being sensitive or not. The statistical determination may be based on any number of users or employees who mark do not mark information as being sensitive. It will be appreciated that the sensitive data engine **212** may relate certain employees higher than others based on their role within the organization, user rights, and/or the like.

In some embodiments, the secure content system **106** may utilize the security rules (e.g., as instructions) to identify sensitive information, redact/replace the sensitive information, generate secure link, recreate an email or document, determine authentication to receive/access the sensitive information, and/or authenticate based on challenges to the secure link.

In some embodiments, security rules may redact and create a security link to replace words, parts of words, phrases, sentences, paragraphs, sections, and/or the like that are identified as sensitive by one or more security rules. In one example where email is referenced, but that applies to any other form of electronic communication, the communication engine **202** may receive an email from a user of an enterprise. The communication engine **202**, the controller engine **204**, or the sensitive data engine **212** may determine which security policy or policies to retrieve based on the email (e.g., based on the sender of the email, the email server that provided the email, group or department of the sending user, recipient of the email, per department of the recipient, whether the recipient is internal to the organization or external, whether the email is to go between departments or outside of the enterprise network, or the like). The content scanning engine **206** may scan contents of the email and/or the email attachments if any for sensitive information utilizing one or more rules of the one or more policies. If sensitive information is found, the sensitive data engine **212** may redact, replace, and/or store the sensitive information. The security link generator **216** may generate a secure link and one or more security challenge(s) based on the one or more rules of the one or more policies.

In various embodiments, security rules may create a security link to replace any number of sections, paragraphs, sentences, phrases, words, or characters that contain one or more words that are identified as sensitive by one or more security rules. In various embodiments, security rules may redact and create a security link to replace the contents of an entire electronic communication (e.g., email, attachment, document, or file) that contains one or more words that are identified as sensitive by one or more security rules. In some embodiments, security rules will redact/replace the entire contents of an email or attachment if it matches a template of an account maintenance communication, or contain keywords associated with the account maintenance communication. The account maintenance communication may include password reset request email or an account change notification email. This is further described herein.

The security rule datastore **210** may be any structure and/or structures suitable for storing data entries or records (e.g., an active database, a relational database, a self-referential database, a table, a matrix, an array, a flat file, a documented-orientated storage system, a nonrelational NoSQL system, an FTS-management system such as Lucene/

Solar, and the like). The security rule data store may store security rules and/or security policies containing security rules.

As discussed herein, it will be appreciated that the secure content system **106** can support any number of enterprise systems. As such there may be any number of security policies and associated security rules for different enterprises. In some embodiments each enterprise will have its own security policies containing its own security rules. It will be appreciated that the secure content system **106** may define a default set of security rules for any number of enterprises or organizations within an enterprise.

In various embodiments the secure content system **106** may learn keywords and/or phrases that indicate sensitive information for different departments within an organization. The secure content system **106** may recommend keywords and/or phrases to different enterprises based on similarity of function, similarity of industry, similarity of sensitive information, or the like. For example information regarding board meetings such as minutes of board meetings, accounting information, categories of employee information, salaries, health information, passwords, and/or the like may be considered as indications of sensitive information across enterprises and organizations within those enterprises. As such the secure content system **106** may recommend keywords and phrases that indicate such categories of information as being potentially sensitive to any number of enterprises and/or organizations within enterprises.

In some embodiments, the sensitive data engine **212** is configured to evaluate content using the one or more security rules in the security category selected by the security policy engine **208** and identify the sensitive data within the content. Security rules may be based on metadata, for example, a security rule may recognize financial statements as sensitive data by looking for keywords such as “quarterly,” “financial,” “subtotal,” and the like.

In various embodiments, the sensitive data engine **212** may apply any analytics depending on the content of an email, attachment, document, or file. For example, the sensitive data engine **212** may identify a category of sensitivity such as “accounting” and detect a number of words or phrases associated with accounting. The sensitive data engine **212** may identify each word and/or phrase associated with accounting, determine proximity of the words and/or phrases considered to be sensitive, evaluate metadata, assess structure and format of the content and/or sensitive words and/or phrases, assess the number of words and/or phrases, and the like to generate a confidence score. The confidence score may be compared to a predetermined threshold (e.g., defined by the enterprise or the secure content system **106**) to determine if any amount or all of the content is sensitive. It will be appreciated that there may be different predetermined thresholds and different scoring functions may be utilized based on different categories of sensitivity (e.g., indicating different analytics, scoring functions, and/or thresholds for different categories including, for example, accounting, legal issues, finance, personal information, trade secrets, or the like). The categories of sensitivity, analytics to be used, scoring functions, and/or thresholds may be defined in the security policy.

Once sensitive data has been identified it may be stored within secure storage **214** (e.g., as vaulted electronic communications, described in further detail with reference to FIG. 17). In some embodiments, security rules may evaluate the content for account maintenance communications such as a password reset request email. In various embodiments, security rules may evaluate the sender or recipients of the

email to determine if an email should be designed as sensitive, or contains sensitive data. For example, a security rule may identify an email as sensitive if a particular person in the enterprise system **120** is the sender or one of a plurality of recipients. In some embodiments, sensitive data engine **212** may be configured to perform textual recognition processes on the contents of the email to identify sensitive data. In various embodiments, the contents includes digital images, digital video and the sensitive data engine **212** may perform image recognition or pattern recognition processes to identify sensitive data.

Secure storage **214** may be any structure and/or structures suitable for storing data entries or records (e.g., an active database, a relational database, a self-referential database, a table, a matrix, an array, a flat file, a documented-orientated storage system, a nonrelational No-SQL system, an FTS-management system such as Lucene/Solar, and the like). The secure storage **214** may be encrypted. It will be appreciated that the secure storage **214** may include any number of storage devices and/or storage networks. Sensitive information may be stored in various different locations and encrypted by any number of encryption keys and or security methodologies.

While the secure storage **214** may be depicted in secure content system **106**, it will be appreciated that in some embodiments there may not be a secure storage **214**. For example, the secure content system **106** may store the sensitive information of a particular enterprise in storage resident in or owned by that particular enterprise. For example, a document containing sensitive information may be retrieved or accessed by the secure content system **106** to determine if that particular document contains sensitive information based on security rules of security profiles. A copy of the document containing a secure link that replaces the sensitive information may be provided back to the document management system and/or to a user. The original document may remain in its original location or be moved to a secure location by the enterprise or by the secure content system **106** depending on embodiments.

In another example, the sender of an email may retain the original email with the original contents in their “sent” file of their enterprise email server. The secure content system **106** may receive the email and transmission to the recipient, scan the email for sensitive information, generate a new email with a secure link in place of the sensitive information, and provide the new email including the secure link to the intended recipient. The secure content system **106**, instead of storing the sensitive information within the secure storage **214**, may associate the sensitive information with the original email in the enterprise server and/or the sender’s “sent” email box. This example is generalizable to any electronic repository that contains any type of electronic communication.

In some embodiments, the security link generator **216** may be configured to generate a security link which links to the secure storage **214** (or the location of the sensitive information) where one of a plurality sensitive data identified by the sensitive data engine **212** is located or is otherwise accessible.

In some embodiments, the security link generator **216** generates different security links for different sensitive data (e.g., different keywords and/or phrases) in the email. For example, different keywords and phrases may be associated with highly sensitive confidential information while other keywords and phrases may be associated with less sensitive confidential information. It will be appreciated that, in some embodiments, the security rules may require different levels

of challenges and therefore different levels of security functions depending on the type of keywords and phrases (e.g., categories of keywords and phrases associated with different levels of confidentiality).

In some embodiments, if there is a particular electronic communication containing keywords and phrases associated with highly sensitive confidential information and other keywords and phrases associated with less sensitive confidential information, the security link generator **216** generates a single security link using security rules associated with the highly sensitive confidential information. In various embodiments, in the case of the particular electronic communication containing keywords and phrases of different levels of confidentiality, the security link generator **216** generates a different security link for each of the different levels of confidentiality.

In some embodiments, the security link is linked to one or more security functions that must be satisfied before a requester can access the sensitive data. In one example the security link is a hyperlink or other executable code that presents a user (e.g., an email recipient or document recipient) with a challenge before being able to access the sensitive information. The challenge, for example, may be a password request which requests a password from the user before allowing the user to access the sensitive information. The security link may require any amount of information. For example, the security link may require that the user be logged into an enterprise system, utilizing a digital device behind an enterprise firewall, utilize a digital device with a particular domain name or MAC address, have an active VPN connection, be a member of a particular domain or group, possess one or more encryption keys, be able to provide one or more codes from other devices in the user's possession (e.g., two factor authentication), and/or any combination of the above.

In various embodiments, the security link may link the user to a part of the secure content system **106** and/or a part of the enterprise system. The secure content system **106** and/or the enterprise system may select and/or create one or more challenges based on the security link (e.g., based on an ID number associated with the security link), the security rules, the type of confidential information, and/or any other criteria.

The security function may be location or time-based. For example, the security function may be satisfied if the secure engine **220** determines that the requester is currently at the location of the company offices during office hours (e.g., based on GPS sensor readings of a digital device of the requester) at the time the request is received. A requester is any user or digital device that activates the security link.

As discussed herein, the security function may provide a request to the requester to enter a password or answer a security question selected by the sender of the email, a representative of the sender of the email, administrator, document creator, file creator, attachment creator, or a user with the appropriate rights. In some examples, the security function may require biometric authentication such as facial recognition, fingerprint, voiceprint, or some combination thereof.

In various embodiments, the security link generator **216** may generate a security link that replaces the entire contents or part of an electronic communication (e.g., email, attachment, document, and/or file). The security link may comprise one security function which must be satisfied before any requester can access the sensitive data associated with the security link. In some embodiments, the security link may comprise a plurality of security functions, the plurality of

security functions may be from a same level of security. In some embodiments, as long as one of the plurality of security functions is satisfied, the requester can access the sensitive data associated with the security link. In various embodiments, all security functions or a combination of security functions must be satisfied before the requester may access the sensitive data.

In some embodiments, replacement engine **218** generates a replacement electronic communication (e.g., attachment, document, or file which includes the security link generated by the security link generator **216**). For example, the replacement engine **218** replaces or redacts the sensitive information with the security link. The replacement engine **218** may replace or redact content containing sensitive information (e.g., containing any number of keywords or phrases known as sensitive based on the security rules, containing a number of keywords or phrases above a particular threshold defined by the security rules, and/or containing one or more particularly sensitive keywords or phrases relative to other keywords or other content based on the security rules), and/or the predetermined number of characters, words, phrases, sentences, paragraphs, sections, or the like in front and/or behind each keyword.

It will be appreciated that the replacement engine **218** may replace different portions of content based on different security rules. For example based on the sensitive information and or physician related to the sensitive information, different security rules may replace different amounts of information. In some embodiments, the replacement engine **218** may replace the same amount of information. For example the replacement engine **218** may replace every sentence or paragraph containing keywords or phrases identified by the security rules.

In some embodiments, after the replacement engine **218** replaced confidential information from an electronic communication, the communication engine **208** may send the replacement electronic communication to a recipient user or otherwise store the electronic communication to an account of the recipient user. The replacement electronic communication may not include the sensitive data identified by the sensitive data engine **212**. In various embodiments, the replacement electronic communication includes additional information regarding the suspicious content, such as a warning of a potential phishing scam. In some embodiments, replacement engine generates a replacement content such as an image, spreadsheet, video, etc.

In various embodiments, the replacement engine **218** may also include additional information such as an explanation that sensitive information was replaced and instructions on how to access the sensitive information. For example, the replacement engine **218** may provide instructions on how to engage the security link and provide information to meet requirements for authentication or security challenges. In another example, the replacement engine **218** may provide instructions for the recipient to contact an administrator or a sender to resolve problems or challenges related to the sensitive information and/or security functions.

In some embodiments, the security engine **220** is configured to receive a request to access the sensitive data from the requester interacting with the security link in the replacement electronic communication. As discussed herein, a requester is a person who receives or attempts to access an electronic communication (e.g., document, file, or email), and interacts with a security link.

In some embodiments, the security engine **220** receives a response to the security function from the requester, and determines if the security function is satisfied. Providing that

the security function is satisfied, the security engine 220 may send a request to the sensitive data provider engine 222 for the sensitive data related to the satisfied security function or otherwise enables the recipient access to the sensitive information.

The security engine 220 may determine if the security function is satisfied in any number of ways. In one example, the security engine 220 may receive a password from the recipient and determine if it matches the expected value in the security function. In another example, the security engine 220 may confirm an encryption key, code, MAC address, biometric value, and/or any other information provided by the requester or the requester's device. In a further example, the security engine 220 may confirm the location of the requester or location of the requester's device. The security engine 220 may also confirm the time of day and/or the date that the request was received and or the document, file, email, or security link was created or provided.

It will be appreciated that the security engine 220 may also provide additional information or additional challenges to the recipient. If the recipient does not sufficiently answer a challenge, provides insufficient information, or does not provides an answer during a predetermined time period, the security engine 220 may be configured to provide additional challenges or provide messages or other alerts to administrators of the failed attempt. In some embodiments the security engine 220 may refuse to accept any other attempts at satisfying security functions if the recipient fails to satisfy one or more security challenges.

In various embodiments, the security engine 220 determines the appropriate security challenge and/or determines the sufficient answer to the security challenge based on security rules and/or security profiles.

In various embodiments, the security engine 220 may be configured to trigger a "lock down" on a user email account, a cloud storage account, or an enterprise email account and the like, if the user is notified of an account maintenance communication, such as a password reset request, and the user indicates that the user did not initiate the maintenance communication. In some embodiments, the presence of a software plug-in, in an internet browser of the user system 102-1 for example, may satisfy the security function.

In some embodiments, the sensitive data provider engine 222 is configured to provide or receive requests or data between the secure storage 214, the security engine 220 and any other component of system 100. The sensitive data provider engine 222 may receive a request from security engine 220 for sensitive data, providing that the security function is satisfied. The sensitive data provider engine 222 may, in turn, sends a request to the secure storage 214 for the sensitive data related to the satisfied security function.

The invitation engine 224 may be configured to send requests to user systems or enterprise systems to subscribe to the secure content system 106. In some embodiments, the invitation engine 224 is configured to invite email recipients or senders to join the secure content system 106. For example, an employee of an enterprise system may request a common recipient of the employees email to be more secure in their email communications. As such, the employee may control the invitation engine 224 to send the recipient an invitation. The invitation engine 224 may generate an invitation and associate that invitation with the recipient and/or a unique identifier. The unique identifier may identify the enterprise and/or the employee. If the recipient accepts the invitation and invitation acceptance is sent to the invitation engine 224. The invitation engine 224 may then add the recipient's email address and any other

information regarding the recipient (e.g., including for example the recipient's name, MAC address, encryption keys, passwords, logins, and the like).

In some embodiments, after the invitation has been accepted, email from the recipient to the enterprise may be intercepted by the secure content system 106. In one example, the recipient (now a sender of email to the employee) may send the email directly to the secure content system 106 or email that is sent by the recipient to the enterprise system may be redirected by the enterprise system to the secure content system 106. Subsequently, the secure content system 106 may identify the email from the recipient, identify the correct security policies, scan the email for sensitive information, generated new email with replaced sensitive information and a security link, and send the new email to the employee. This example, while written with respect to email, is generalizable to any electronic communication format (e.g., general chat messages like slack messages, and any other format consonant with any form of "electronic communication" defined herein).

In some embodiments, after the invitation has been accepted email from the employee to the recipient may no longer be scan for sensitive information may be scan for only a subset of sensitive information depending on the policy rules.

It will be appreciated that, in some embodiments, unless the recipient accepts the invitation, the secure content system 106 may, in some instances, not scan email for sensitive information and replace sensitive information with a link in order not to disrupt business. In some embodiments, an employee may request secure content system 106 not replace sensitive information (or scan for sensitive information) of a particular email thread, attachment, document, or file for a limited time. In one example, this may allow the employee to have multiple revisions and discussions without having to overcome recurring challenges either by himself or by the recipient. In some embodiments, certain employees with specific authorization as defined by the security rules and or security policy may have the option to suspend all or some services of secure content system 106 (including for example not scanning certain documents, email, or files for sensitive information or replacing sensitive information in those documents, emails, or files with secure links).

Content datastore 226 may be any structure and/or structures suitable for storing data entries or records (e.g., an active database, a relational database, a self-referential database, a table, a matrix, an array, a flat file, a documented-orientated storage system, a nonrelational No-SQL system, an FTS-management system such as Lucene/Solar, and the like). Content from different users or enterprise associated with secure content system 106 may be stored in different partitions of and/or different locations associated with the content datastore 226, so that if an email account of the user system 102-1 is hacked, content associated with the enterprise DMS 126 will not be affected.

In some embodiments, the replacement electronic communication generated by the replacement engine 218 is stored in the content datastore 226, and in various embodiments, after the replacement electronic communication is generated by the replacement engine 218, the electronic communication may be deleted from the third-party email server 110 or the enterprise email system 124. In some embodiments, using email as an exemplary electronic communication, all of the email from a user's email account on the third-party email server, such as AOL mail may be deleted from the third-party email server and transferred to the content datastore 226. In some embodiments, people

who send email to the user's cloud or other email services (e.g., AOL mail), may be seamlessly forwarded to the content datastore **226**. The user will be able to send and receive email using the AOL mail but still be able to keep their email and other content more secure than on the AOL mail server.

Notification engine **228** is configured to notify the user of changes in the user's content or accounts. In some embodiments, the secure content system **106** can provide additional services based on content scanned the email or document. For example, a common malicious attack on an email recipient may be a phishing scam requesting personal information, credit card numbers, logins, passwords, and the like. Another common malicious attack, may be a phishing or spear phishing attack requesting the user to provide personal information or to change a password on an email account that the user may have even though the email is not the user's email system. The secure content system **106**, upon determining this type of email content, may notify the recipient that this is a common attack or a likely malicious email.

In some embodiments, the notification engine **228** sends SMS or MMS messages to the user to determine if the user initiated an account maintenance communication (e.g., an email requesting a change of password or confirmation of other personal information). In one example, the notification engine **228** may send the message to the user's personal mobile device or any other digital device to allow the user to be notified in two different ways and/or on two different devices for security. In various embodiments, the notification engine **228** informs a sender of an email with sensitive data that one or more recipients of the email have successfully (or unsuccessfully) accessed the sensitive data. For example, the notification engine **228** notifies the user by using email, text messaging, telephone call, and/or other forms of notification.

It will be appreciated that content may include any kind of digital content or media, including, but not limited to pictures, text, video, sound, graphics, icons, interactive programming, or any combination of the above. All e-mail specific examples herein are for illustrative purposes only, and are generalizable to any electronic communication format.

Search module **229** enables a search to be performed on all vaulted communications associated with a user after passing a single challenge. Details of search module **229** are described in further detail below with reference to FIGS. **16-18**.

FIG. **3** depicts a flowchart **300** of the process of securing contents in some embodiments. In step **302**, the communication engine **202** facilitates a request from sender user system **102-1** to send an email from the sender user system **102-1**. The secure content system **106** may monitor the sender user system **102-1**'s email communications on the enterprise email system **124**. For example, in some embodiments the enterprise email system **124** may receive email from the sender user system **102-1** and provide that email to the communication engine **202**. In some embodiments, the user system **102-1** (e.g., a browser, email application, or other application) may be configured to provide the email to the secure content system **106** before the enterprise email system **124** receives the email.

In step **304**, the content scanning engine **206** of the secure content system **106** may be configured to evaluate the email from the enterprise email system **124**. In one example, the content scanning engine **206** may be configured to identify the correct security policies and/or security rules based on

information of the email (e.g., based on the sender, recipient, user system **102-1**, identifier in the email, metadata associated with the email, MAC address, or any other information). The content scanning engine **206** makes scan the contents of the email or sensitive information based on the security policies and/or security rules.

In step **306**, the sensitive data engine **212** may be configured to evaluate the contents of the email based on the security rules to identify sensitive data or suspicious data. In some embodiments, the security policy engine **208** may be configured to select one or more security categories to apply to the content scanned by the content scanning engine **206**. Once a security category has been selected, security rules associated with the selected security category stored in the security rule datastore **210** may be used by the sensitive data engine **212** to evaluate the contents of the email to identify sensitive data. The user of sender user system **102-1** may designate an email as sensitive, such as the email designating interface **810** in FIG. **8**. The content may be contained within an email and may include textual components. In some embodiments, the content scanning engine **206** may perform textual recognition processes on the contents of the email. In some embodiments, the content includes digital images, digital video and the content scanning engine **206** may perform image recognition or pattern recognition processes.

In step **308**, the sensitive data identified by the sensitive data engine **212** may be stored in secure storage **214**. In various embodiments, the sensitive data identified by the sensitive data engine **212** may be stored in any location such as but not limited to the enterprise system.

In step **310**, the security link generator **216** may be configured to generate a security link that must be satisfied in order to access the sensitive data. In some embodiments the security link may point to the location (e.g., memory address) within secure storage **214** where one of a plurality sensitive data identified by the sensitive data engine **212** is stored. The security link may be associated with one or more security function(s) that must be satisfied before a requester (e.g., a user or device that interacts with the security link) can access the sensitive data. In some embodiments, the requester is the recipient of the email, or a representative of the recipient of the email. The security link may be included in a replacement email generated by replacement engine **218**.

Although termed as a "replacement email," it will be appreciated that all or part of the email may be replaced with a security link and the original email may be directed to the recipient. In some embodiments the original email is replaced with a similar email but lacking the sensitive information and including the security link before sending onto the recipient (or the recipient's mailbox).

In some embodiments, the replacement email does not include the sensitive data identified by the sensitive data engine **212**, but would include a secure link which contains a security function. The security function issues a challenge to the requester, the challenge presented to the user may require the requester to enter a password or answer a security question selected by the sender of the email, or a representative of the sender of the email. In some embodiments, the sender of the email determines the type of security function presented to subsequent recipients of the email. In various embodiments, the enterprise system **120** determines the type of security function presented to subsequent recipients of the email according to a level of sensitivity of the email.

In step **312**, the communication engine **202** facilitates a request to send the replacement email generated by the

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replacement engine **218** to the recipient user system **102-2**. In the above example, contents secured by the secure content system **106** is an email. In various embodiments, the contents may include digital images, digital videos.

The secure content system **106** may proactively scan any number of emails and any email storage system to identify secure or sensitive information. The secure content system **106** may develop or update security rules and/or security profiles based on the scan. In some embodiments, the secure content system **106** may replace the sensitive information and any other information based on security rules with a secure link. In one example the secure content system **106** may store the sensitive information elsewhere. As a result, email stored in the email server, email mailbox, or the like may include copies of email without the sensitive information. If a malicious user gains access to the email server or email mailbox, then the malicious user would still have to overcome the challenge of the security link. It will be appreciated that any number of mailboxes or accounts may be scanned and sensitive information removed from any number of emails. In some embodiments, some mailboxes or accounts may be scanned and sensitive information withdrawn while other mailboxes or accounts may not be scanned, depending on the owner of the mailbox, security rules, or any other reason.

In various embodiments, the secure content system **106** may store the sensitive information and/or any scanned information to enable for faster searching of both sensitive and non-sensitive information contained in any number of email, documents, or other files. For example, the secure content system **106** may store all or parts of the scanned information in a database or other data structure to enable fast searching of information. In this way the user may be able to search through all of their email, documents, or other files, in search of information without concern that sensitive information may not be found during a search.

It will be appreciated that the secure content system **106** is not limited to email or other messages, but may provide access and security to documents and other files. For example, in step **302**, the communication engine **202** may receive a request to access a document stored in a document management system (e.g., enterprise DMS **126**) or other storage. The secure content system **106** may monitor communications with the enterprise DMS **126** and/or the enterprise DMS **126** (or other device) may route requests for documents or files (or the documents or files themselves) through the secure content system **106**.

In step **304**, the content scanning engine **206** of the secure content system **106** may be configured to evaluate the document or file. In one example, the content scanning engine **206** may be configured to identify the correct security policies and/or security rules based on information of the document, information of the file, or the request (e.g., based on the requester, the DMS, the enterprise, the type of content of the document or file, or any other information). The content scanning engine **206** makes scan the contents of the document or file for sensitive information based on the security policies and/or security rules.

In step **306**, the sensitive data engine **212** may be configured to evaluate the contents of the document or file based on the security rules to identify sensitive data or suspicious data. In some embodiments, the security policy engine **208** may be configured to select one or more security categories to apply to the content scanned by the content scanning engine **206**. Once a security category has been selected, security rules associated with the selected security category stored in the security rule datastore **210** may be used by the

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sensitive data engine **212** to evaluate the contents of the document or file to identify sensitive data.

In various embodiments, a document administrator, secretary, or user that initially stored the document or file may designate an email as sensitive. The content may be contained within an document or file and may include textual components. In some embodiments, the content scanning engine **206** may perform textual recognition processes on the contents of the document or file. In some embodiments, the content includes digital images, digital video and the content scanning engine **206** may perform image recognition or pattern recognition processes.

In optional step **308**, the sensitive data identified by the sensitive data engine **212** may be stored in secure storage **214**. In various embodiments, the sensitive data identified by the sensitive data engine **212** may be stored in any location such as but not limited to the enterprise system.

In step **310**, the security link generator **216** may be configured to generate a security link that must be satisfied in order to access the sensitive data. In some embodiments the security link may point to the location (e.g., memory address) within secure storage **214** where one of a plurality sensitive data identified by the sensitive data engine **212** is stored. The security link may be associated with one or more security function(s) that must be satisfied before a requester (e.g., a user or device that interacts with the security link) can access the sensitive data. In some embodiments, the requester is the recipient of the email, or a representative of the recipient of the email. The security link may be included in a replacement document or file generated by replacement engine **218**.

Although termed as a "replacement document or file," it will be appreciated that all or part of the document or file may be replaced with a security link and the original document or file may be directed to the user seeking to access the document or file. In some embodiments the original document or file is replaced with a similar document or file but lacking the sensitive information and including the security link before sending onto the recipient (or the recipient's mailbox).

In some embodiments, the replacement document or file does not include the sensitive data identified by the sensitive data engine **212**. The security function issues a challenge to the requester, the challenge presented to the user may require the requester to enter a password or answer a security question (e.g., selected by the author or user that stored the document or file). In some embodiments, the author or user that stored the of the document or file determines the type of security function presented to subsequent recipients of the document or file. In various embodiments, the enterprise system **120** determines the type of security function presented to subsequent recipients of the document or file according to a level of sensitivity of the document or file (or content).

In step **312**, the communication engine **202** facilitates a request to send the replacement document or file generated by the replacement engine **218** to the recipient user system **102-2**. In the above example, contents secured by the secure content system **106** is an document or file. In various embodiments, the contents may include digital images, digital videos.

FIG. **4** depicts a flowchart **400** of an example process of challenging a requester before retrieving sensitive data from the secure storage **214**. In step **402**, the secure content system **106** may receive a request from the recipient user system **102-2** that received an email with security link to access sensitive data. The email may include the security

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link generated by the security link generator **216**. The security engine **220** may receive a request to access the sensitive data from the requester interacting with the security link in the replacement email. In some embodiments, upon receiving the request from the recipient user system **102-2**, the notification engine **228** may send a notification to the sender user system **102-1**, such as the notification interface **910** from FIG. **9**.

In step **404**, the security link generated by the security link generator **216** includes a security function which issues a challenge to a requester. The security function must be satisfied before the requester can access the sensitive data. In various embodiments, the security link is associated with executable code that produces a challenge for the recipient. In some embodiments the security link is linked to a security challenge that is remote from the recipient (e.g., the security link triggers a challenge or provides information from the secure content system **106**). The challenge may be a password, a code, or any other information. In some embodiments the security link may not request any information from the requester but rather may retrieve information from the requester's digital device (e.g., VPN identifiers, MAC addresses, encryption keys, encryption key identifiers, and the like).

In step **406**, the security engine **220** receives a response to the security function. For example, the security engine **220** may receive a password, code, MAC address, identifier, encryption key, and/or the like. In various embodiments, it will be appreciated the any digital device may receive the response to the security function and/or perform authentication. In some embodiments, a digital device of the user includes an application or executable code on a web browser that may perform security functions (e.g., determine time, location, MAC address, encryption keys, and the like) and/or authenticate the user or the user's digital device based on the security functions.

In step **408**, the security engine **220** determines if the security function is satisfied. As discussed herein, any digital device may receive the response or result of the security functions and may authenticate the information. The security function may be chosen by the enterprise employing the service. Different security functions may be employed to access different sensitive information. For example, if the sensitive information is categorized as being highly confidential (e.g., hiring/firing decisions, salaries, strategies, core technologies, trade secrets, and the like), the enterprise may employ multiple security functions (e.g., two factor authentication and encryption keys) while other sensitive information may be categorized as being less confidential and, as a result, the enterprise may require less robust security functions or fewer security functions (e.g., a password). In various embodiments, the enterprise may configure the security functions required as well as the appropriate responses to the security challenges.

In step **410**, if the security engine **220** determines that the security function is satisfied, the security engine **220** may send a request to the sensitive data provider engine **222** for the sensitive data associated with the satisfied security link. The sensitive data provider engine **222** receives the request from the security engine **220** and may enable access to the secure storage **214** for the sensitive data related to the satisfied security function.

The security engine **220** may enable access to the sensitive data in any number of ways. In various embodiments, the security engine may provide the sensitive data to the requester or provide a link whereby the requester may view the sensitive data on a remote server (e.g., the secure content

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system **106** or the enterprise system) without being able to download and/or copy the information. Alternately, the security engine **220** may provide the sensitive information in any form to the requester or provide instructions to any other digital device to allow the requester access to the sensitive information.

In various embodiments, the security engine **220** may not allow access to the sensitive data by others if they are not in the correct location, it is the wrong time of day, the email that was to include the sensitive data was sent too long in the past (e.g., beyond a threshold amount of time or days and therefore access has expired), a request to lock the sensitive information is received (e.g., from the email sender or an employee with sufficient data rights), there is an indication that the sender's account has been targeted, and/or the like. Similarly, in various embodiments, the security link may also become nonfunctional and/or no answer to the challenge will be deemed to satisfy the challenge if the recipient is not in the correct location, it is the wrong time of day, the email that was to include the sensitive data was sent too long in the past (e.g., beyond a threshold amount of time or days and therefore access has expired), a request to lock the sensitive information is received (e.g., from the email sender or an employee with sufficient data rights), there is an indication that the sender's account has been targeted, and/or the like.

It will be appreciated that the secure content system **106** is not limited to email or other messages, but may provide access and security to documents and other files. For example, in step **402**, the secure content system **106** may receive a request from the recipient user system **102-2** that received a document or file with security link to access sensitive data. The document or file may include the security link generated by the security link generator **216**. The security engine **220** may receive a request to access the sensitive data from the requester interacting with the security link in the replacement document or file. In some embodiments, upon receiving the request from the recipient user system **102-2**, the notification engine **228** may send a notification to a digital device of the document or file author, moderator, or administrator.

In step **404**, the security link generated by the security link generator **216** includes a security function which issues a challenge to a requester. The security function must be satisfied before the requester can access the sensitive data.

In step **406**, the security engine **220** receives a response to the security function. For example, the security engine **220** may receive a password, code, MAC address, identifier, encryption key, and/or the like. In various embodiments, it will be appreciated the any digital device may receive the response to the security function and/or perform authentication. In some embodiments, a digital device of the user includes an application or executable code on a web browser that may perform security functions (e.g., determine time, location, MAC address, encryption keys, and the like) and/or authenticate the user or the user's digital device based on the security functions.

In step **408**, the security engine **220** determines if the security function is satisfied. As discussed herein, any digital device may receive the response or result of the security functions and may authenticate the information. The security function may be chosen by the enterprise employing the service. Different security functions may be employed to access different sensitive information. For example, if the sensitive information is categorized as being highly confidential (e.g., hiring/firing decisions, salaries, strategies, core technologies, trade secrets, and the like), the enterprise may

employ multiple security functions (e.g., two factor authentication and encryption keys) while other sensitive information may be categorized as being less confidential and, as a result, the enterprise may require less robust security functions or fewer security functions (e.g., a password). In various embodiments, the enterprise may configure the security functions required as well as the appropriate responses to the security challenges.

In step 410, if the security engine 220 determines that the security function is satisfied, the security engine 220 may send a request to the sensitive data provider engine 222 for the sensitive data associated with the satisfied security link. The sensitive data provider engine 222 receives the request from the security engine 220 and may enable access to the secure storage 214 for the sensitive data related to the satisfied security function.

The security engine 220 may enable access to the sensitive data in any number of ways. In various embodiments, the security engine may provide the sensitive data to the requester or provide a link whereby the requester may view the sensitive data on a remote server (e.g., the secure content system 106 or the enterprise system) without being able to download and/or copy the information. Alternately, the security engine 220 may provide the sensitive information in any form to the requester or provide instructions to any other digital device to allow the requester access to the sensitive information.

In various embodiments, the security engine 220 may not allow access to the sensitive data by others if they are not in the correct location, it is the wrong time of day, the email that was to include the sensitive data was sent too long in the past (e.g., beyond a threshold amount of time or days and therefore access has expired), a request to lock the sensitive information is received (e.g., from the email sender or an employee with sufficient data rights), there is an indication that the sender's account has been targeted, and/or the like. Similarly, in various embodiments, the security link may also become nonfunctional and/or no answer to the challenge will be deemed to satisfy the challenge if the recipient is not in the correct location, it is the wrong time of day, the email that was to include the sensitive data was sent too long in the past (e.g., beyond a threshold amount of time or days and therefore access has expired), a request to lock the sensitive information is received (e.g., from the email sender or an employee with sufficient data rights), there is an indication that the sender's account has been targeted, and/or the like.

FIG. 5 depicts a flowchart 500 of the process of securing contents in some embodiments. In step 502, the security policy engine 208 is configured to generate security rules. Each security rule is associated with at least one criteria. In some embodiments, security rules may be associated with the sender or the recipient, for example, the entire contents of an email sent by a corporate attorney of an enterprise may be automatically marked as sensitive. In various embodiments, security rules may be associated with the location or time that a requester is trying to access the sensitive data by interacting with the security link. In one example security rules may be associated with a particular enterprise.

In step 504, the security policy engine 208 organizes a plurality rules into a security category according to the criterion associated with each of the plurality of security rules. In some embodiments, a security rule may be organized into one or more security categories. The plurality of security rules may be stored in the security rule datastore 210. Security categories may organize security rules according to industry (e.g., such as healthcare, military, banking, or

the like), group, organization, department, function, and/or organizational culture (e.g., such as temporary employees, external contractors, interns, executive, or the like).

In step 506, the communication engine 202 facilitates a request from sender user system 102-1 to send an email from the sender user system 102-1. The secure content system 106 may monitor the sender user system 102-1's email communications on the enterprise email system 124. In some embodiments, the content scanning engine 206 of the secure content system 106 may be configured to evaluate the email from the enterprise email system 124.

In step 508, the security policy engine 208 may be configured to select one or more security categories to apply to the content scanned by the content scanning engine 206 based on the content that was scanned, the email sender, the email receiver, location, time of day, metadata associated with the email, attachments, scanned content of attachments (which may also contain sensitive information), and/or any other information.

In step 510, subsequent to step 508 in which based on one or more security category has been selected the security rules associated with the selected security category stored in the security rule datastore 210 may be used by the sensitive data engine 212 to evaluate the contents of the email to identify sensitive data. The content may be contained within an email (and/or attachment of the email) and may include textual components. In some embodiments, the content scanning engine 206 may perform textual recognition processes on the contents of the email. In some embodiments, the content includes digital images, digital video and the content scanning engine 206 may perform image recognition or pattern recognition processes.

In step 512, the security link generator 216 may be configured to generate a security link which enables execution of a local function (e.g., local to the entity that executed the link such as the recipient's computer), enables execution of any number of security functions on any number of digital devices to challenge the requester (i.e., the requester being the user or digital device that executed or interacted with the security link). The security link includes at least one security function that is satisfied before a requester can access the sensitive data. In some embodiments, the requester is the recipient of the email, or a representative of the recipient of the email. The security link may be included in the replacement email generated by replacement engine 218.

In some embodiments, the replacement email does not include the sensitive data identified by the sensitive data engine 212. The security function issues a challenge to the requester, the challenge presented to the user may require the requester to enter a password or answer a security question selected by the sender of the email, or a representative of the sender of the email. In some embodiments, the sender of the email determines the type of security function presented to subsequent recipients of the email. In various embodiments, the enterprise system 120 determines the type of security function presented to subsequent recipients of the email according to a level of sensitivity of the email.

In the above example, contents secured by the secure content system 106 is an email. In various embodiments, the contents may include digital images, digital videos.

As discussed herein, systems and methods described may be utilized in the context of documents and/or files. For example, in step 506, the communication engine 202 facilitates a request from a user or user digital device to access a document or file. The secure content system 106 may monitor the sender user system 102-1's request for the document or file. In some embodiments, the secure content

system **106** may monitor or intercept requests for documents or files from a file server or document server (or any digital devices). In some embodiments, the content scanning engine **206** of the secure content system **106** may be configured to evaluate the requested document(s) or file(s) from the enterprise email system **124**.

In step **508**, the security policy engine **208** may be configured to select one or more security categories to apply to the content scanned by the content scanning engine **206** based on the content that was scanned, the document(s) and/or file(s) requester, the DMS/file server storing the documents or files, categories or metadata associated with the document(s) and/or file(s), location of the document(s) and/or file(s), time of day, and/or any other information.

In step **510**, subsequent to step **508** in which based on one or more security category has been selected the security rules associated with the selected security category stored in the security rule datastore **210** may be used by the sensitive data engine **212** to evaluate the contents of the document(s) and/or file(s) to identify sensitive data. In some embodiments, the content scanning engine **206** may perform textual recognition processes on the contents of the document(s) and/or file(s). In some embodiments, the content includes digital images, digital video and the content scanning engine **206** may perform image recognition or pattern recognition processes.

In step **512**, the security link generator **216** may be configured to generate a security link which enables execution of a local function (e.g., local to the entity that executed the link such as the recipient's computer), enables execution of any number of security functions on any number of digital devices to challenge the requester (i.e., the requester being the user or digital device that executed or interacted with the security link). The security link includes at least one security function that is satisfied before a requester can access the sensitive data. In some embodiments, the requester is the requester of the document(s) and/or file(s), or a creator/administrator/editor/modifier of one or more of the document(s) and/or file(s). The security link may be included in replacement document(s) and/or file(s) generated by replacement engine **218**.

In some embodiments, the replacement document(s) and/or file(s) do not include the sensitive data identified by the sensitive data engine **212**. The security function issues a challenge to the requester, the challenge presented to the user may require the requester to enter a password or answer a security question selected by an administrator or user associated with the document(s) and/or file(s). In some embodiments, the administrator or user associated with the document(s) and/or file(s) determines the type of security function presented to those that wish to access one or more of the document(s) and/or file(s). In various embodiments, the enterprise system **120** determines the type of security function presented to subsequent requesters of the document(s) and/or file(s) according to a level of sensitivity of the document(s) and/or file(s).

FIG. 6 depicts a flowchart **600** of the process of account maintenance authentication according to some embodiments. In step **602**, the content scanning engine **206** of the secure content system **106** scans the content on a media platform such as an email account of the recipient user system **102-2** in the third-party email server **110** of FIG. 1. The secure content system **106** may be configured to monitor the email communications of the recipient user system **102-2** on the third-party email system **110** by evaluating the content based on security rules. The sensitive data engine **212** may evaluate the contents of the email account to

identify sensitive data within the content. The sensitive data engine **212** may evaluate the content using security rules which may redact any amount of contents (e.g., including entire contents) of an email if it matches a template of an account maintenance communication, or contain keywords associated with the account maintenance communication. The account maintenance communication may include password reset request email or an account change notification email.

In step **604**, when the sensitive data engine **212** identifies sensitive data, in the form of the account maintenance communication and transfers data from the account maintenance communication to the secure storage **214**. In some embodiments, the data transferred from the account maintenance communication to secure storage **214** comprise the URL of the internet address to reset a password, information such as the user name, address of the recipient user system **102-2**.

In step **606**, the notification engine **228** notifies the user of recipient user system **102-2** of changes in the user's contents or accounts or, alternately, requests for such changes. In some embodiments, the notification engine sends SMS or MMS messages to the user to determine if the user initiated an account maintenance communication. In various embodiments, the notification engine inform a sender of an email with sensitive data that one or more recipients of the email have successfully (or unsuccessfully) accessed the sensitive data. A communication interface **1000** from FIG. 10 shows an example notification of a plurality of password reset requests. Notifications **1010**, **1020**, **1030**, and **1040**, which may be display on the output device **1514** from FIG. 15, show the notification from four different web applications or websites of account maintenance communications. In some embodiments, the notification engine **228** informs the user of changes in the user's contents or accounts by sending a notification email to an alternate email address of the user, such as example notification email **1200** from FIG. 12.

In step **608**, if the user of recipient user system **102-2** recognizes the account maintenance communication from the notification engine **228**. The user may send an authenticate message to the secure content system **106** recognizing that the account maintenance communication is known to the user. The secure content system **106** receives the authenticate message and as a result, controller engine **204** may send a request to the secure storage **214** for the data transferred from the account maintenance communication to the secure storage **214**. For example, the user may send an authentication message by interacting with the "Reset Securely" electronic button **1032** from FIG. 10.

In step **610**, if the user of recipient user system **102-2** does not recognize the account maintenance communication from the notification engine **228**. The user may send a decline message to the secure content system **106**, informing the secure content system **106** of a possible breach of the user's email account. The secure content system **106** may trigger a content lockdown of the user's email account. In some embodiments, the secure content system **106** may log an instance of the recipient user system **102-2**'s rejection of the account maintenance communication and the secure content system **106** does not trigger an account lockdown or content lockdown associated with the user's email account until the number of rejections from the user and/or other users that received similar messages exceeds a rejection threshold value. The content lockdown may comprise suspending the sending outgoing emails and the reception of incoming emails. The user of recipient user system **102-2** may be notified of the lockdown of the user's email account by the

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notification engine 228. For example, the user may send a decline message by interacting with the “Didn’t Request” electronic button 1034 from FIG. 10. In some embodiments, if the user of recipient user system 102-2 does not recognize the account maintenance communication from the notification engine 228, the notification engine 228 may send an alert to the recipient user system 102-2 using an alternate method of communication such as SMS, MMS or an alternate email address of the user of recipient user system 102-2.

FIG. 7 depicts a flowchart 700 of the process of suspicious content mitigation according to some embodiments. In step 702, the content scanning engine 206 of the secure content system 106 may scan or evaluate the email from the enterprise email system 124 or the third-party email server 110.

In step 704, the sensitive data engine 212 may evaluate the contents of the email (and/or evaluate the contents of any number of attachments of the email) based on security rules to identify suspicious content. The suspicious content may be a phishing email and security rules may identify keywords “account,” “secur,” “auth,” “update,” and the like (including, in some embodiments, synonyms and words with similar meaning). The security rules which may be used to identify suspicious content may include verifying the authenticity of a sender of an email. For example, an email displaying the sender of an email as “Apple Purchase Receipt,” however, further investigation yields that the email address associated with “Apple Purchase Receipt” has no affiliations with Apple. In some embodiments, one of a first plurality of recipients receiving the email from the originating sending may identify suspicious content in the email designate the email suspicious.

In step 706, the security link generator 216 may generate a security link which replaces the identified suspicious content (the link may be within the email and/or within any number of attachments). The generated security link may point to a location (e.g., memory address) within the secure storage 214 where the identified suspicious content is stored. Alternately, the security link may trigger security functions to enable access to the suspicious content. The security link generator 216 may replace some or all of the contents of email (and/or attachments) with an example interface 1110 from FIG. 11. The example interface 1110 may comprise a message informing the sender and recipients of the email of the potential phishing scam.

In step 708, the identified suspicious content may be stored in the secure storage 214, third-party cloud storage, enterprise storage, or any other device(s). By storing the identified suspicious content in the secure storage 214 and/or elsewhere, recipients of the email will not be given the opportunity to interact with the phishing email, reducing the chances of spreading the phishing email.

In step 710, the replacement engine 218 generates a replacement email which includes the security link generated by the security link generator 216 (e.g., in the email and/or attachments). The generated security link may include the example interface 1110 from FIG. 11.

In some embodiments, once an email identified as containing suspicious content, by the secure content system 106 or by the one of the first plurality of recipient, the email containing the suspicious content is replaced by the replacement email in the electronic mailboxes of the remaining first plurality of recipients, and any subsequent recipient of the email subscribed to the secure content system 106. The secure content system 106 may log the instance of the suspicious content in a suspicious report. The suspicious report may be used by other users or enterprises of system 100 to determine future suspicious content.

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FIG. 13 depicts a block diagram of an example computing device 1302 according to some embodiments. Any user system 102, content delivery system 106 and domain system 108 may comprise an instance of computing device 1302. Computing device 1302 comprises a processor 1304, a memory 1306, a storage 1308, an input device 1310, a communication network interface 1312 and an output device 1314. Processor 1304 is configured to execute executable instructions (e.g., programs). In some embodiments, the processor 1304 comprises circuitry or any processor capable of processing the executable instructions.

Memory 1306 stores data. Some examples of memory 1306 include storage devices, such as RAM, ROM, RAM cache, virtual memory, etc. In various embodiments, working data is stored within memory 1306. The data within memory 1306 may be cleared or ultimately transferred to storage 1308.

Storage 1308 includes any storage configured to retrieve and store data. Some examples of storage 1308 includes flash drives, hard drives, optical drives, and/or magnetic tape. Each of memory system 1306 and storage system 1308 comprises a computer-readable medium, which stores instructions or programs executable by processor 1304.

Input device 1310 is any device that inputs data (e.g., mouse, keyboard, stylus). Output device 1314 outputs data (e.g., speaker, display, virtual reality headset). It will be appreciated that storage 1308, input device 1310 and output device 1314 may be optional. For example, routers/switches may comprise processor 1304 and memory 1306 as well as a device to receive and output data (e.g., communication network interface 1312 and/or output device 1314).

Communication network interface 1312 may be coupled to a network (e.g. communication network 104) via communication network interface 1312. Communication network interface 1312 may support communication over an Ethernet connection, a serial connection, a parallel connection, and/or an ATA connection. Communication network interface 1312 may also support wireless communication (e.g., 802.11 a/b/g/n, WiMax, LTE, WiFi). It will be apparent that communication network interface 1312 may support many wired and wireless standards.

FIG. 17 depicts exemplary modules of a search module used to search vaulted information, in accordance with an embodiment. As depicted in FIG. 17, search module 229 includes search request module 1702, vault check module 1704, vault search module 1706, results module 1708, file migration module 1710, and fingerprint module 1712. The modules depicted with reference to search module 229 are merely exemplary; fewer or more modules may be used to achieve the functionality described herein.

The term “vaulted”, as used herein, may refer to a category of electronic communications that are stored in secure storage 214 (as opposed to non-secured or less secure storage). Also within the “vault” may be files that are not electronic communications, but are nonetheless electronic files requested by the user to exist as secure data within secure storage 214. Examples of such files may include redacted attachments from electronic communications, such as images or spreadsheets. The term “files” or “vaulted files”, as used herein with respect to being included within the vault, broadly encompasses electronic communications and any other files that are vaulted. The embodiment of FIG. 17 assumes that electronic communications that are redacted have copies stored, in their original form (that is, without redaction), at secure storage 214, and together form vaulted electronic communications. In some embodiments described above with respect to FIGS. 1-13, accessing vaulted elec-

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tronic communications requires passing a challenge on a per-item basis, where when the challenge is passed, the user gains access to a requested vaulted electronic communication in unredacted form. While these embodiments are effective in unlocking access to vaulted items, a user may have myriad vaulted electronic communications, and without simultaneous access to multiple vaulted items through which to run a search, the user may be unable to identify which electronic communication is relevant to their purposes. Search module 229 solves this limitation by unlocking a plurality of vaulted items for search.

Search request module 1702 receives a request to search a vault from a user of an electronic communication repository 1600. Turning briefly to FIG. 16, FIGS. 16A-16D shows a sequence of user interfaces involved in performing a search through multiple emails within a vault. As depicted in FIG. 16A, electronic repository 1600 may include redacted electronic communications. Electronic repository 1600 may operate using a native application designed specifically to navigate electronic communications stored therein (e.g., an application for navigating an e-mail inbox or an application for navigating various SMS conversations). Searching solely using such a native application for electronic repository 1600 for information that may be redacted from messages therein does not result in the search including results that would have matched had the matching information not been redacted. A user may have multiple electronic repositories 1600, each dedicated to different environments (e.g., one for general chat messages like instant messages in Microsoft teams, one for slack messages, one for SMS messages, one for emails, and so on). Search request module 1702 may offer a request to search through multiple electronic communication repositories. Search request module 1702 may offer selectable options corresponding to each electronic repository available to a user, whereby the user may select a subset of electronic repositories to search (or a subset of vaulted data that was originally addressed to each selected electronic repository to search).

In order to search such redacted information, search request module 1702 enables receipt of a search request for searching within the vault. The request to search the vault may be received by way of receiving a selection from a user of selectable option 1610, which as depicted is a "search vault" option. Selectable option 1610 enables the user to in effect open the vault for the purposes of searching the vault in one shot. That is, rather than having to overcome a challenge to read any individual redacted electronic communication, a search is able to be performed across plural vaulted messages at once after passing a challenge that is prerequisite to the search being performed.

In an embodiment, selectable option 1610 is within a non-native interface relative to the native interface for electronic repository 1600. The non-native interface may be integrated into the native interface by way of an extension to the native interface of the electronic communication repository. For example, a vault application that provides access to the vaulted information may be the non-native application, and may have an application programming interface (API) that enables the vault application to extend functionality of the native application to include selectable option 1610. In an embodiment, the non-native interface may be i-frame of the vault application that is integrated into the native interface, the i-frame including images provided by the vault application in a secure context within the interface of the native application. In an embodiment, the non-native inter-

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face may be displayed in a separate window from a window that renders displays from the native application.

Responsive to receiving a selection of selectable option 1610, vault check module 1704 initiates an identity verification challenge to a device of the user. Challenge mechanisms are described in detail in the foregoing description of FIGS. 1-13, and any above-described challenge may be used as the identity verification challenge here. As depicted in FIG. 16B, a status of a challenge and any other aspects of the challenge may be depicted in challenge interface 1620. Challenge interface 1620 may occur in the non-native interface (e.g., as integrated into the native interface using an interface extension in the manner described above). Vault check module 1704 may determine whether the identity verification challenge is successful, and responsive to determining that the identity verification challenge is successful, may search the vaulted information for one or more files comprising a symbol input by the user. The term symbol, as used herein, may refer to, e.g., a string of text characters, an image, a video, and/or any other content which the user wishes to use as a reference to search for vaulted (e.g., redacted) electronic communications having the symbol.

Following a successful challenge, vault search module 1706 may search vaulted electronic messages for terms that match a symbol input by the user. In an embodiment, the symbol may be input by the user into selectable option 1610 (that is, prior to the challenge being initiated), and responsive to passing the challenge, the symbol may be searched. In an embodiment, turning to FIG. 16C, the symbol may be input into selectable option 1630, which is displayed following a successful challenge. The symbol may be input in any fashion (e.g., by typing text; by dragging-and-dropping images, video, and/or files, selectors, buttons, etc.).

Regardless of how or when the symbol is entered, vault search module 1706 searches for electronic communications that have content matching the symbol. The search may be limited to vaulted electronic communications, or may include matches from the electronic repository of the user (e.g., electronic repository 1600) as well as the vaulted electronic communications. The search may be limited to content that was redacted from original messages, rather than to the entire message, in order to reduce processing power and time required to complete the search by reducing the universe of what needs to be searched. The search may be performed on any redacted information, including text and/or other attachments within original messages, and the search may be performed on unredacted information as well.

After matches are identified from the search, results module 1708 generates for display identifiers corresponding to files having content matching the symbol. For example, as shown in FIG. 14, a search string of a sensitive number results in search results of vaulted files corresponding to that number in results interface 1640. Results interface 1640 may be part of the non-native interface of the vault application, and may operate as an extension to the native interface of the electronic communications repository or may be a stand-alone interface. In keeping the results within results interface 1640, each of the vaulted files remain secure. Results module 1708, responsive to receiving a selection of an identifier of a vaulted file, may display content of the vaulted file within results interface 1640 (e.g., within an i-frame within the native application) to ensure that the vaulted file remains securely vaulted while still available for browsing.

File migration module 1710 may migrate one or more files from the non-native application to the native application in response to receiving a request to do so. For example, the user may request to drag-and-drop a vaulted file to

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electronic repository interface **1600**, or to any other non-secure interface (e.g., a folder in a file explorer system). The user may additionally and/or alternatively provide a file directory address with an address to where the file is to be moved, or may instruct migration in any other fashion. Responsive to a vaulted file being migrated out of the vault, file migration module **1710** may attach a condition to the file to be migrated back to the vault. The condition may be attached to the file, such that the file auto-deletes or auto-migrates back to the vault. For example, the condition may be to delete the file after the file is accessed and then closed. Another example condition may be to delete the file after a certain amount of time has passed since the file was migrated. When a file is migrated, in an embodiment, only a copy of the file is migrated, and a copy of the file remains vaulted. In another embodiment, when the file is migrated, no copy of the file remains on the vault, and instead of deleting the file, the file is moved back to the vault when the condition is reached. In an embodiment, file migration module **1710** directly monitors for the condition being reached, and instructs the system housing the file to delete or otherwise return the file when the condition is reached.

In an embodiment, fingerprint module **1712** fingerprints vaulted files in a manner that uniquely identifies a user who accesses the vaulted file. For example, responsive to detecting that a user is accessing and/or migrating a vaulted file, fingerprint module **1712** may apply a fingerprint to the file. The fingerprint may be used to attribute movement of that file to the user. For example, where a user maliciously accesses a vaulted file to cause harm, fingerprint module **1712** may ingest a copy of the vaulted file and determine the exact user who removed the file from the vault, and may provide the identity of the user to an administrator for further action.

In an embodiment, because users may prefer to avoid having to log into a vault to perform a search, dummy stubs (sometimes referred to as redacted stubs) may be used. Secure content system **106** may generate a dummy stub for an electronic communication when redaction is performed on a message. For example, where an attachment to an electronic communication is deemed to have sensitive data that is to be redacted, secure content system **106** may remove the attachment from the electronic communication, but may leave metadata about the attachment (e.g., title) in the electronic communication. FIG. **14** depicts redacted stubs in an electronic communication and an option to retrieve what was redacted. As shown in electronic communication **1400**, attachments **1410** are redacted, but their file names remain intact, thus showing metadata of a sensitive attachment remaining while the sensitive attachment itself is removed. Therefore, when a user performs a search using native interface **1600**, the file names of sensitive files may be searched (or any other metadata that is not redacted). This results in added functionality, where one does not need to access vaulted files to search such metadata of sensitive files. Responsive to receiving a selection of selectable option **1420** (or going through the sequence of FIGS. **16A-D**), the original message including the sensitive attachments may be retrieved. FIG. **15** depicts unredacted stubs in an unredacted electronic communication following identity verification. Retrieval of the original message **1500** also retrieves the unredacted attachments **1510**, which may be searched when performing a search of vaulted files (e.g., for a symbol matching something in the full attachment).

FIG. **18** depicts an exemplary process for using the search module to search vaulted information, in accordance with an embodiment. Process **1800** may be performed by search

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module **229** executing modules of FIG. **17** using instructions stored on memory **1206** and/or storage **1208** as executed by one or more processors **1204**. Process **1800** begins with search module **229** receiving **1802** (e.g., using search request module **1702**), based on input by a user into an interface of an electronic communication repository (e.g., selectable option **1610** and/or selectable option **1630**), a request to search through vaulted information, the vaulted information redacted from the electronic communication repository.

Search module **229** may then transmit **1804** an identity verification challenge to a device of the user (e.g., using vault check module **1704**). Search module **229** may determine **1806** whether the identity verification challenge is successful and, responsive to determining that the identity verification challenge is successful, may search **1808** the vaulted information for one or more files comprising a symbol input by the user (e.g., using vault search module **1706**). Search module **229** may generate **1810**, for display within the interface, identifiers corresponding to each of the one or more files (e.g., using results module **1708**).

The invention claimed is:

1. A method for searching information redacted from electronic communications, the method comprising:

receiving, based on input by a user into an interface of an electronic communication repository, a request to search through vaulted information, the vaulted information redacted from the electronic communication repository, wherein, as electronic communications are received, the electronic communications are scanned for content satisfying redaction rules, the content satisfying redaction rules is redacted and stored as part of the vaulted information in a separate repository, and the electronic communications are stored in redacted format to the electronic communication repository without the content satisfying redaction rules;

responsive to receiving the request to search through the vaulted information, transmitting an identity verification challenge to a device of the user;

determining whether the identity verification challenge is successful;

responsive to determining that the identity verification challenge is successful, searching the vaulted information in the separate repository for one or more files comprising a symbol input by the user, wherein redacted content from a plurality of electronic communications is searched when searching the vaulted information; and

generating, for display within the interface, identifiers corresponding to each of the one or more files.

2. The method of claim **1**, wherein the interface comprises an extension to a native interface of the electronic communication repository, the extension added by a vault application that provides access to the vaulted information.

3. The method of claim **2**, wherein the interface is an i-frame of the vault application that is integrated into the native interface.

4. The method of claim **1**, wherein the vaulted information comprises an original message, the original message including sensitive data that was redacted from the electronic communication repository.

5. The method of claim **1**, wherein transmitting an identity verification challenge to a device of the user occurs responsive to receiving the request to search through the vaulted information.

6. The method of claim **1**, wherein the symbol is input with the request to search through the vaulted information.

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7. The method of claim 1, further comprising, further responsive to determining that the identity verification challenge is successful:

generating for display a search field; and
receiving, based on search input in the search field, the symbol.

8. The method of claim 1, wherein the vaulted information is stored separately from unvaulted information stored in the electronic communication repository.

9. The method of claim 8, wherein searching the vaulted information comprises additionally searching the unvaulted information, and wherein the identifiers comprise a given identifier that identifies at least one file within the unvaulted information.

10. The method of claim 9, wherein the unvaulted information comprises a dummy stub having searchable metadata corresponding to redacted sensitive data, and wherein the given identifier is part of the identifiers based on the symbol matching the searchable metadata.

11. The method of claim 1, wherein the interface is a non-native interface through which the vaulted information can be accessed securely, and wherein the method further comprises:

receiving a selection of a given identifier for a vaulted file; and
securely displaying, through the interface, contents of the vaulted file without migrating the vaulted file to the electronic communication repository.

12. The method of claim 1, wherein the interface is a non-native interface through which the vaulted information can be accessed securely, and wherein the method further comprises:

receiving a selection of a given identifier for a vaulted file; and
adding a copy of the vaulted file to the electronic communication repository.

13. The method of claim 12, further comprising deleting the copy of the vaulted file from the electronic communication repository responsive to detecting that the copy of the vaulted file was accessed.

14. The method of claim 12, further comprising deleting the copy of the vaulted file from the electronic communication repository responsive to determining that a predetermined amount of time has elapsed from a time at which the copy of the vaulted file was added to the electronic communication repository.

15. The method of claim 1, wherein a file of the one or more files is fingerprinted with a fingerprint that stays with the file when copies of the file is transmitted to other users, and wherein the user is attributed to the copies of the file based on the fingerprint.

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16. A non-transitory computer-readable medium comprising memory with instructions encoded thereon for searching information redacted from electronic communications, the instructions, when executed, causing one or more processors to perform operations, the instructions comprising instructions to:

receive, based on input by a user into an interface of an electronic communication repository, a request to search through vaulted information, the vaulted information redacted from the electronic communication repository, wherein, as electronic communications are received, the electronic communications are scanned for content satisfying redaction rules, the content satisfying redaction rules is redacted and stored as part of the vaulted information in a separate repository, and the electronic communications are stored in redacted format to the electronic communication repository without the content satisfying redaction rules;

responsive to receiving the request to search through the vaulted information, transmit an identity verification challenge to a device of the user;

determine whether the identity verification challenge is successful;

responsive to determining that the identity verification challenge is successful, search the vaulted information in the separate repository for one or more files comprising a symbol input by the user, wherein redacted content from a plurality of electronic communications is searched when searching the vaulted information; and

generate, for display within the interface, identifiers corresponding to each of the one or more files.

17. The non-transitory computer-readable medium of claim 16, wherein the interface comprises an extension to a native interface of the electronic communication repository, the extension added by a vault application that provides access to the vaulted information.

18. The non-transitory computer-readable medium of claim 17, wherein the interface is an i-frame of the vault application that is integrated into the native interface.

19. The non-transitory computer-readable medium of claim 16, wherein the vaulted information comprises an original message, the original message including sensitive data that was redacted from the electronic communication repository.

20. The non-transitory computer-readable medium of claim 16, wherein transmitting an identity verification challenge to a device of the user occurs responsive to receiving the request to search through the vaulted information.

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